

A binomial count of meromorphic travelling waves

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Abstract

We study meromorphic travelling waves of $u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0$, a constant-coefficient family containing KdV–Burgers, Kuramoto–Sivashinsky and Kawahara. After integration and normalization, the profiles satisfy $P(D)v + v^2/2 = 0$, where $D = d/dz$ and P is monic of degree $p \geq 2$ with complex coefficients.

For evolution equations of even order (p odd), and for purely dispersive equations (only odd-order derivatives), we give a complete algebraic description of all nonconstant globally meromorphic travelling waves: they are rational, rational in an exponential, or elliptic, with one pole per period.

One-pole waves of these three kinds are exceptional: for $p \geq 3$ a generic equation of the family has none. We count the exceptions as pairs (P, v) , over the whole family: there are $\binom{2p-1}{p-1}$ with v rational in an exponential of given rate, and $2\binom{2p-2}{p-2}$ with v elliptic on a given generic lattice. We also count the solitary pulses of the purely dispersive subfamily and the fronts of the purely dissipative one (only even-order derivatives in the linear part).

When $2p-1$ is prime, these statements are proved and all pairs are distinct. We conjecture them in every order.

Keywords. Meromorphic travelling waves; dispersive and dissipative equations; elliptic functions; polynomial convolution; algebraic multiplicity.

1 Introduction

Exact travelling waves of nonlinear evolution equations are usually found one equation at a time, and usually only for special values of the coefficients. The generalized Kuramoto–Sivashinsky equation $u_t + uu_x + u_{xx} + \sigma u_{xxx} + u_{xxx} = 0$ is a classical example. It has nonconstant travelling waves that are polynomials in a hyperbolic tangent for exactly four values of the dispersion coefficient, up to its sign: $\sigma = 0, 4, 12/\sqrt{47}$ and $16/\sqrt{73}$. Kuramoto and Tsuzuki found the wave of the dispersionless equation [39], Kudryashov the others [26, 27], and Eremenko proved that, together with Kudryashov’s elliptic solution and a single rational one, they exhaust the nonconstant meromorphic travelling waves [18]. The classical tables list six cases at these four values [5, 33]. They do not distinguish a wave from its mirror image, which solves the equation with the opposite sign of σ . If the two are counted separately, the two cases with $\sigma = 0$ are their own mirror images and the other four are not, so the list has $2 + 2 \cdot 4 = 10$ members. This paper shows that ten is $\binom{5}{2}$: the classical list is the case $p = 3$ of a binomial enumeration, proved for infinitely many p and conjectured for all.

We consider the evolution equations

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0,$$

which contain the KdV–Burgers ($p = 2$), Kuramoto–Sivashinsky ($p = 3$), Kawahara and Benney–Lin ($p = 4$) and Nikolaevskiy ($p = 5$) equations; Table 1 lists the named members with $p \leq 6$. A travelling wave $u = w(x - ct)$ satisfies, after one integration, $\sum_j b_j w^{(j)} - cw + w^2/2 + K = 0$. Shifting w by a constant solution and normalizing the leading coefficient gives

$$E_P(v) := P(D)v + \frac{1}{2}v^2 = 0, \quad P(\rho) = \rho^p + \sum_{j=0}^{p-1} a_j \rho^j, \quad p \geq 2, \quad (1.1)$$

with $D = d/dz$ and complex coefficients. The coefficients $a_1, \dots, a_{p-1}$ are those of the evolution equation, up to a rescaling of x , and the speed, the integration constant and the background are absorbed into a_0 (Section 2). Throughout, p is the order of the profile equation (1.1), which is one less than the order of the evolution equation: Kuramoto–Sivashinsky is a fourth-order equation and has $p = 3$, Kawahara has $p = 4$, and the seventh-order KdV equation has $p = 6$. Phrases such as “order three” or “in every order” always refer to p .

The waves. A travelling wave is *meromorphic* if its profile is a single-valued meromorphic function of the complex variable z . The familiar exact solutions are of this kind: a polynomial in $\tanh$ or sech is a rational function of an exponential $e^{\kappa z}$, a cnoidal wave is an elliptic function, and rational functions of z also occur. These are the three kinds of wave in this paper. We consider those with *one pole per period*: all the poles of the profile are translates of a single pole by its periods. For a wave of $\tanh$ type the period in question is the imaginary period $2\pi i/\kappa$ of $e^{\kappa z}$, and the rate κ measures the width of the pulse or front. For many equations of the family, described below, these are all the meromorphic travelling waves that exist.

The question. For $p \geq 3$ such waves are exceptional: an equation of the family has one only if its coefficients satisfy algebraic relations, as the four values of σ illustrate. This suggests the right question. We do not ask how many waves a given equation has, but how many pairs of an equation and a wave there are in the whole family.

To make this question finite, two continuous freedoms must be removed. A translate of a wave is again a wave, so we place the pole at a fixed point. Rescaling x changes the width of a wave and the coefficients of its equation together, so we fix the width: for a wave that is rational in an exponential we prescribe the rate $\kappa = 1$, and for an elliptic wave we prescribe both of its periods, that is, its period lattice. The coefficients of the equation remain free. We therefore count pairs (P, v) , where P ranges over all monic polynomials of degree p and v solves (1.1):

- a *rational-exponential pair* has v rational in e^z , with one pole modulo $2\pi i\mathbb{Z}$, placed at $i\pi$, and $v \rightarrow 0$ as $e^z \rightarrow 0$;
- an *elliptic pair* has v elliptic on a given lattice, with its only pole at zero.

One more convention explains a factor two below. The profile equation has two constant solutions, and a wave can be measured from either of them; this changes P but not the wave (Section 2). A rational-exponential wave is measured from its limit as $e^z \rightarrow 0$. An elliptic wave has no such limit, and both choices are counted, so every elliptic wave appears in two pairs.

The answer. The numbers of pairs are

$$N_p = \binom{2p-1}{p-1} \quad \text{rational-exponential pairs,}$$

$$M_p = 2 \binom{2p-2}{p-2} \quad \text{elliptic pairs on a generic lattice.}$$

We call $2p-1$ the *index* of the order p . Our main result establishes these numbers whenever the index is a prime number:

- there are exactly N_p rational-exponential pairs, and they are all distinct (Theorem 5.4);
- exactly $S_p = \binom{2p-2}{p-2}$ of them are *pulses*, which return to the same constant at both ends, and the others are *fronts*, which connect two different constants (Proposition 5.6);
- a generic lattice carries exactly $M_p = 2S_p$ elliptic pairs, and they are all distinct (Theorem 6.1).

The index is prime for $p = 2, 3, 4, 6, 7, 9, 10, 12, \dots$. At the three remaining orders up to twelve, $p = 5, 8$ and 11 , the same statements are established by exact computer calculations (Corollary 7.2). We call the orders with a prime index, together with all $p \leq 12$, the *covered* orders; they include every named equation of Table 1. We conjecture the same statements in every order (Conjecture 10.1).

Table 1: The named equations of the family with $p \leq 6$; here $u_{kx} = \partial_x^k u$. Each name stands for a family with free coefficients, and the numbers count distinct pairs across that family: rational-exponential pairs at unit rate, and elliptic pairs on a generic given lattice, where every elliptic wave appears twice. An equation with only odd-order, or only even-order, linear terms belongs to a pure subfamily, counted by F_p and $2F_p$ (see below); the full families are counted by N_p and M_p and contain the pure subfamily of the same p , so the three pairs of KdV–Burgers include the KdV soliton. Except in the two rows marked $\dagger$, these pairs are all the nonconstant meromorphic travelling waves of the equations, apart from a rational wave when all lower coefficients vanish.

Equation	Linear terms	p	Rational-exponential pairs	Elliptic pairs
KdV	u_{3x}	2	$F_2 = 1$	$2F_2 = 2$
KdV–Burgers †	u_{2x}, u_{3x}	2	$N_2 = 3$	$M_2 = 2$
Kuramoto–Sivashinsky (KS)	u_{2x}, u_{4x}	3	$F_3 = 2$	0
KS with dispersion	u_{2x}, u_{3x}, u_{4x}	3	$N_3 = 10$	$M_3 = 8$
Kawahara	u_{3x}, u_{5x}	4	$F_4 = 3$	$2F_4 = 6$
Benney–Lin †	$u_{2x}, \dots, u_{5x}$	4	$N_4 = 35$	$M_4 = 30$
Nikolaevskiy	u_{2x}, u_{4x}, u_{6x}	5	$F_5 = 6$	0
Nikolaevskiy with dispersion	$u_{2x}, \dots, u_{6x}$	5	$N_5 = 126$	$M_5 = 112$
seventh-order KdV	u_{3x}, u_{5x}, u_{7x}	6	$F_6 = 10$	$2F_6 = 20$

Table 1 lists the counts for the named equations of the family with $p \leq 6$: the Benney–Lin equation [2, 40], the Kuramoto–Sivashinsky equation with dispersion, also called the Benney or KdV–KS equation [24], the Nikolaevskiy equation [44] and its dispersive version [32, 46], the long-wave reductions of Kudryashov et al. [34], and the equations already mentioned. The small

entries are familiar. The three pairs of KdV–Burgers are the KdV soliton and the KdV–Burgers front, read in its two directions. The two elliptic pairs of KdV are the cnoidal wave on its two backgrounds. The eight elliptic pairs at $p = 3$ are Kudryashov’s elliptic solution: on a given lattice it occurs for four values of the dispersion coefficient, each time on two backgrounds (Section 9.2).

Four remarks make precise what the numbers mean. The counts are over $\mathbb{C}$ and range over all monic P . Many of the pairs have real coefficients and give smooth bounded waves of real equations: eight of the ten at $p = 3$, and 21 of the 35 at $p = 4$; Section 9 identifies them. The word *distinct* carries content: the pairs are the solutions of a polynomial system with as many equations as unknowns, and a solution of such a system can be multiple, like a multiple root of a polynomial. On special lattices elliptic pairs do merge in this way, and Section 9.2 determines where this happens at order three. Rational waves are not counted: at the covered orders the only one that vanishes at infinity is $v = c_* z^{-p}$, for $P = D^p$. Finally, the waves of low order are classical, and Section 9 gives the references; what is new is that the lists are complete, and how long they are. We do not discuss stability, and we treat only the quadratic nonlinearity uu_x .

Every wave has a label. Here is where the binomial coefficient N_p comes from. Put $Q = 1/(1 + e^{-z}) = (1 + \tanh(z/2))/2$. Every rational-exponential pair has a profile of the form $v_A = s_p A(D)Q$, where s_p is a fixed scale and A is a monic polynomial of degree $p - 1$; this is a polynomial in $\tanh(z/2)$ written in a convenient basis. For $\operatorname{Re} z < 0$ we have $Q = e^z - e^{2z} + e^{3z} - \dots$, and D multiplies e^{nz} by n . Hence the coefficient of e^{nz} is $\pm s_p A(n)$ in v_A and $\pm s_p P(n)A(n)$ in $P(D)v_A$, while in v_A^2 it is $\pm s_p^2 S_A(n)$ with the convolution sum

$$S_A(n) = \sum_{j=1}^{n-1} A(j)A(n-j).$$

These sums are the values of a polynomial S_A of degree $2p - 1$, and (1.1) holds exactly when $2P(n)A(n) = s_p S_A(n)$ for all n . So the pair exists if and only if the polynomial A divides the polynomial S_A , and then P is the quotient, up to the constant factor (Proposition 5.1).

The smallest case shows what happens. For $p = 2$ and $A = \rho + a$,

$$S_A(\rho) = (\rho - 1) \left(\frac{\rho(\rho + 1)}{6} + a\rho + a^2 \right),$$

which is divisible by $\rho + a$ exactly when $a(a - 1)(a + 1) = 0$. The solution $a = 0$ is the KdV soliton, and $a = \pm 1$ are the KdV–Burgers front read in its two directions. Now reduce modulo $3 = 2p - 1$. Whatever the value of a ,

$$3S_A(\rho) \equiv -(\rho^3 - \rho) = -\rho(\rho - 1)(\rho + 1) \pmod{3},$$

so modulo 3 the divisor A must be one of the three factors ρ , $\rho - 1$, $\rho + 1$. Each of them comes from exactly one solution, and the root of A modulo 3 serves as a label: $\{0\}$ for the soliton, $\{2\}$ and $\{1\}$ for the two fronts.

The same happens whenever $\ell = 2p - 1$ is prime. The polynomial S_A is a combination of the binomial coefficients $\binom{\rho-1}{k}$, and only the last of them, with $k = \ell$, has ℓ in its denominator. For this reason ℓS_A reduces modulo ℓ to $\pm(\rho^\ell - \rho)$, whatever A is, and the condition becomes that A divides $\rho^\ell - \rho$ modulo ℓ . Now $\rho^\ell - \rho$ is the product of $\rho - s$ over all residues s modulo ℓ . Its monic divisors of degree $p - 1$ are therefore given by the $(p - 1)$ -element subsets S of the integers modulo ℓ , and there are $\binom{\ell}{p-1} = N_p$ of them. We prove that every pair can be reduced modulo ℓ ,

although its coefficients are in general algebraic numbers and not integers, and that every divisor comes from exactly one pair. For instance $A = \rho^2 + 1/11$, the Kuramoto–Tsuzuki wave, reduces to $\rho^2 + 1 = (\rho - 2)(\rho - 3)$ modulo 5, and has the label $\{2, 3\}$.

So each pair has a label S , and the labels carry information. The pair is a pulse exactly when S contains 0, and the mirror image of the pair with label S has label $-S$. At $p = 3$ the ten classical waves correspond to the ten two-element subsets of $\{0, 1, 2, 3, 4\}$ (Table 2), and the four pulses are the subsets that contain 0.

The elliptic count comes from the pulses. When one period of the lattice tends to infinity, an elliptic pair becomes a pulse, measured from one of its two constant backgrounds. We show that every pulse, on either background, arises in this way exactly once, and this gives $M_p = 2S_p$.

Purely dispersive and purely dissipative equations. In the evolution equation, derivatives of odd order are dispersive terms and derivatives of even order are dissipative terms. We call an equation *purely dispersive* if it contains only odd-order derivatives, as the KdV and Kawahara equations do; then p is even and P is an even polynomial. We call it *purely dissipative* if its linear part contains only even-order derivatives, as for the Kuramoto–Sivashinsky equation without dispersion; then p is odd and $P - P(0)$ is an odd polynomial. All other equations, such as KdV–Burgers, are *mixed*.

The two pure subfamilies are singled out by reflection. Reading a wave in the opposite direction, $z \mapsto -z$, gives a wave of the mirror-image equation, in which the terms of one parity have changed sign. For a front the zero limit is then at the wrong end, so one also changes the background; the result is again a pair, the *mirror image* of the first (Section 5.4). The pairs that coincide with their mirror image are exactly those whose equation is purely dispersive, for even p , or purely dissipative, for odd p (Theorem 5.8). In terms of labels they are the pairs with $S = -S$, and the number of such subsets is

$$F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}.$$

At the covered orders this is the number of rational-exponential pairs in the pure subfamily of order p , and the purely dispersive equations also account for $2F_p$ of the elliptic pairs on a generic lattice (Corollary 6.2).

The two subfamilies carry different kinds of rational-exponential waves. Those of a purely dispersive equation are all pulses, at every even p . Those of a purely dissipative equation are all fronts, at the covered orders; for waves with real coefficients this holds at every odd p . For example, the Kawahara family has three rational-exponential pairs and six elliptic pairs on a generic lattice, which are F_4 and $2F_4$; the seventh-order KdV family has ten pairs, two of them real; and the six pairs of the purely dissipative family with $p = 5$ contain the four kink solutions of Kudryashov and Migita [32] (Section 9).

When these are all the meromorphic waves. In two cases the one-pole waves of the three kinds are all the nonconstant meromorphic travelling waves there are (Theorem 3.1). The first case is p odd, that is, every evolution equation of even order. The second is the purely dispersive case. The proof rests on Eremenko’s theorem [18]: an autonomous algebraic differential equation has only these three kinds of meromorphic solutions if it has a single monomial of top degree, here $v^2/2$, and finitely many Laurent expansions at a pole. For odd p the Laurent expansion at a pole is unique. For a purely dispersive equation one Laurent coefficient is free, a resonance;

but the equation is invariant under $z \mapsto -z$ and conserves an energy, and fixing the energy fixes that coefficient. In both cases every nonconstant meromorphic travelling wave has one pole per period, and for a given operator P , which records the equation and the speed, there is at most one such profile up to translation when p is odd, and at most one on each energy level when the equation is purely dispersive.

Mixed operators with p even, such as that of KdV–Burgers, are not covered by this theorem, and cannot be: Section 10 recalls an example with $p = 2$ that has other meromorphic waves. For these operators we still count the one-pole waves of the three kinds, without claiming that there are no others; they are the rows marked $\dagger$ in Table 1. Section 4 writes each of the three kinds as a finite polynomial system, following the principal-part construction of Demina and Kudryashov [15, 16].

A single equation. A count across the family is not a count for each individual equation. To recover the waves of a given equation, we restore the length scale and retain only the pairs whose coefficients match it. Changing the scale traces a curve in the space of coefficients, and at the covered orders the equations with a one-pole rational-exponential wave are those on finitely many such curves, at most one for each pair and its mirror image. This is the precise sense in which the waves are exceptional, and it explains the four values of σ . It refers to the full family: a pure subfamily with few coefficients, such as the KdV or Kawahara equations, can lie entirely on these curves, and at $p = 2$ every equation does, since every KdV–Burgers equation has its front. If at least one lower-derivative coefficient is nonzero, an equation has only finitely many wave shapes after translations and Galilean shifts are identified; the scale-invariant case is different, and the KdV equation, for example, retains its family of solitons of all widths. Section 8 makes this precise and recovers the four Kuramoto–Sivashinsky curves.

For equations of even order the classification gives more. A generic such equation has no nonconstant meromorphic travelling wave at all; this is proved for $3 \leq p \leq 573$ (Corollary 8.3). It may well have travelling waves; but none of them is meromorphic, and in particular none is given by a formula of the kinds above. Here generic means outside a proper algebraic subset of the coefficients.

Section 8 also records two consequences for the way exact travelling waves are found in practice. Many other equations, with mixed derivatives or in several space variables, have the same profile equation, so that the counts apply to them. And the expansion methods by which such waves are usually sought, in powers of $\tanh$ or of a Jacobi function, can only produce waves that are counted here; for a named equation the counts are a checklist for their output.

Other orders. The proofs use the prime ℓ in two places, and these are the two facts that are not yet proved in every order. The first is *rigidity*: the only rational solution of (1.1) with one pole that vanishes at infinity is $c_* z^{-p}$, with $P = D^p$. Equivalently, the only monic polynomial A of degree $p - 1$ that divides its continuous self-convolution $\int_0^\rho A(t)A(\rho - t) dt$ is ρ^{p-1} . As the coefficients of the equation vary, a pair could disappear only by moving off to infinity, and rigidity is the statement that this does not happen; it gives all the counts when pairs are counted with multiplicity. It is proved when $2p - 1$ is a prime or a prime power, and established by exact computations for $p \leq 573$. The second fact is that every rational-exponential pair is simple, which makes the pairs distinct. It is proved when $2p - 1$ is prime and, by exact computations, at $p = 5, 8$ and 11 . Both are conjectured in general. Section 7 states what follows from each of them, and a companion note in the source package contains the proofs of those statements.

Relation to earlier work. The classification theorem is a uniform version of results that exist in pieces. The case of odd p is contained in Yuan et al. [49, Theorem 1], and the energy device for even P is that of Eremenko et al. [19, Lemmas 3–4], which has been applied to the Kawahara, generalized Bretherton and seventh-order KdV equations; Section 3 gives the detailed attributions. The polynomial systems are instances of the principal-part construction of Demina and Kudryashov [15, 16]. The general enumeration has no precedent that we know of; its special cases at $p = 3$ and in the Kawahara family are classical, as the examples make explicit.

Organization. Section 2 contains the normalizations, and Sections 3 and 4 the classification and the three polynomial systems. Section 5 proves the count of rational-exponential pairs at a prime index and reads the labels, and Section 6 proves the elliptic count. Section 7 records what is known at the other orders, and Section 8 reads the count for a single equation. Section 9 works out the lowest orders and the two pure subfamilies, including the lattices on which elliptic pairs merge. Section 10 collects the open questions, and Appendix A describes the exact verification and the ancillary material.

2 Equations, waves and what is counted

This section fixes the normalizations and the vocabulary used in the rest of the paper. The normalizations are what makes a finite count possible, and they are also what connects the count with a given evolution equation.

2.1 From the evolution equation to the profile equation

Let $u = w(\xi)$, $\xi = x - ct$, be a travelling wave of

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0. \quad (2.1)$$

One integration gives

$$\sum_{j=1}^p b_j w^{(j)} - cw + \frac{1}{2}w^2 + K = 0 \quad (2.2)$$

with a constant K . Let w_* be a constant solution of (2.2), that is, a root of $w^2/2 - cw + K$, and let $\lambda \neq 0$. The substitution

$$w(\xi) = w_* + b_p \lambda^p v(\lambda \xi) \quad (2.3)$$

turns (2.2) into the profile equation (1.1) with

$$a_j = \frac{b_j}{b_p} \lambda^{j-p} \quad (1 \leq j \leq p-1), \quad a_0 = \frac{w_* - c}{b_p \lambda^p}. \quad (2.4)$$

Thus $a_1, \dots, a_{p-1}$ record the equation, up to the scale λ , and a_0 records the speed of the wave relative to the background w_* . The scale is free at this stage. It will be fixed by the period of the wave.

The constant solutions of (1.1) are $v = 0$ and $v = -2a_0$. They correspond to the two possible choices of the background w_* . Changing that choice replaces the pair (P, v) by

$$(P - 2P(0), v + 2P(0)), \quad (2.5)$$

which again satisfies (1.1), as a direct substitution shows. After the original background is restored, the two pairs describe the same wave of the same evolution equation. They coincide exactly when $a_0 = 0$.

2.2 Dispersive, dissipative and mixed equations

In (2.1) a derivative of odd order is a dispersive term, and a derivative of even order is a dissipative term: for real coefficients the first changes the phase of a Fourier mode and the second its amplitude. We use the following names.

- The equation is *purely dispersive* if it contains only odd-order derivatives, that is, $b_j = 0$ for odd j . Then p is even, P is an even polynomial, and the equation is invariant under $(x, t) \mapsto (-x, -t)$. Examples are the KdV equation ($p = 2$), the Kawahara equation ($p = 4$) and the seventh-order KdV equation ($p = 6$).
- The equation is *purely dissipative* if its linear part contains only even-order derivatives, that is, $b_j = 0$ for even j . Then p is odd, $P - P(0)$ is an odd polynomial, and the equation is invariant under $(x, u) \mapsto (-x, -u)$. Examples are the Kuramoto–Sivashinsky equation without dispersion ($p = 3$) and the Nikolaevskiy equation ($p = 5$); see Simbawa et al. [46].
- All other equations are *mixed*. Examples are the KdV–Burgers equation and the Kuramoto–Sivashinsky equation with dispersion.

The two pure subfamilies are described by one condition. We call a monic polynomial P of degree p *reflection-symmetric* if $P - P(0)$ has the parity of ρ^p :

$$P(-\rho) = P(\rho) \quad (p \text{ even}), \quad P(\rho) + P(-\rho) = 2P(0) \quad (p \text{ odd}). \quad (2.6)$$

In the travelling-wave variable both symmetries are the reflection $z \mapsto -z$. If P is even and $v(z)$ solves (1.1), so does $v(-z)$. If p is odd and $P - P(0)$ is odd, so does $-v(-z) - 2P(0)$; this is the reflection combined with the change of background (2.5).

2.3 Waves, poles, pulses and fronts

Throughout, a meromorphic function is single-valued and meromorphic on the whole complex plane. Three kinds of meromorphic travelling waves occur in this paper: rational in z , rational in an exponential $e^{\kappa z}$, and elliptic. For the equations covered by Theorem 3.1 there are no others; for the remaining equations these are the waves that we count. The period group of such a wave is its *full* group of periods: trivial in the first case, infinite cyclic in the second, a lattice in the third. We say that the wave has *one pole per period* if its poles form a single coset of this group. There is then one pole in the rational case, one pole modulo the primitive period in the second case, and one pole in each period parallelogram in the elliptic case.

A wave that is rational in $e^{\kappa z}$ and has one pole per period has finite limits as $e^{\kappa z} \rightarrow 0$ and ∞ (Section 4). Both limits are constant solutions. The wave is a *pulse* if the two limits coincide and a *front* otherwise. For a real rate and a real profile these are the solitary waves and the kinks of the evolution equation, and they are smooth and bounded on the real axis when the pole lies off it. In (2.3) we take w_* to be the limit as $e^{\kappa z} \rightarrow 0$. Then $v \rightarrow 0$ at that end, and the other limit is 0 for a pulse and $-2a_0$ for a front.

2.4 What is counted

We count pairs (P, v) : a monic polynomial P of degree p and a solution v of (1.1) of a given kind, with its periods fixed.

- A *rational-exponential pair* has v rational in e^z , with one pole modulo $2\pi i\mathbb{Z}$, placed at $i\pi$, and $v \rightarrow 0$ as $e^z \rightarrow 0$.
- An *elliptic pair* on a lattice Λ has v elliptic with period lattice Λ , including its scale, and its only pole modulo Λ at zero.

Fixing the period or the period lattice removes the scale λ in (2.3), and placing the pole removes translations. The operator P varies over *all* monic polynomials of degree p , and all counts are over $\mathbb{C}$. A wave and its mirror image are counted as two pairs when they are distinct; the pairs that coincide with their mirror image are the subject of Section 5.4. In the elliptic case both pairs (2.5) are counted; in the rational-exponential case the condition $v \rightarrow 0$ selects one of them. Section 8 translates the counts back to a single evolution equation.

3 The classification theorem

This section proves Theorem 3.1 below: for every evolution equation of even order (p odd), and for every purely dispersive equation (P even), a nonconstant meromorphic solution of (1.1) is rational, rational in an exponential, or elliptic, and has one pole per period. The proof has three steps. We first compute the Laurent expansion at a pole and show that for odd p it is unique. For even P it is not: one coefficient is left free, and we show that the energy, a first integral of (1.1), fixes it, so that Eremenko's theorem applies on each energy level. Finally we show that any two poles of one solution differ by a period.

Let $\mathcal{W}$ consist of rational functions of z , rational functions of $e^{\kappa z}$ with $\kappa \neq 0$, and elliptic functions. Period groups are full period groups, as in Section 2; for a rational function of $e^{\kappa z}$ we choose κ so that $2\pi i/\kappa$ generates the group. For the general analytic background, see Eremenko [17].

Theorem 3.1 below collects the analytic input to the count. It is a uniform version of results that exist in pieces. For odd p the indicial polynomial has no positive integer zero, the Laurent expansion at a pole is unique, and Eremenko's theorem applies directly; this case is contained in Yuan et al. [49, Theorem 1], with related results for odd p in Huang et al. [21], Yuan et al. [50] and a fifth-order instance in Dang [10]. For even p the Laurent recurrence has a resonance at $n = 3p$, where one coefficient is left free, and the conclusion need not hold for mixed operators: the Fisher equation has an elliptic-exponential family of solutions [1, 6, 30] whose generic members are meromorphic and of none of the three kinds. The familiar Ablowitz–Zeppetella front is a rational-exponential degeneration of this family. When P is an even polynomial the equation has a first integral, the energy, and fixing its value removes the resonant freedom. This is the device of Eremenko et al. [19, Lemmas 3–4] for $P = D^p + a_0$, stated as a general step by Conte et al. [7, 8] and applied to the Kawahara equation [14, 48], the generalized Bretherton equation [15, 38] and the seventh-order KdV equation [9]. Theorem 3.1 extends it uniformly to every even P , intermediate derivatives included. Equations with two dominant monomials, or without the finiteness property (finitely many Laurent expansions at a pole), are treated by other methods in Demina [12, 13]. We give the proofs in full because Sections 4 and 5 use the Laurent recurrence and the energy dependence in explicit form.

Theorem 3.1 (Classification). *Assume either that p is odd or that P is an even polynomial; that is, the evolution equation has even order or is purely dispersive. Every nonconstant meromorphic solution of (1.1) belongs to $\mathcal{W}$. Every pole has order p , and the pole set is one coset of the full period group: the solution has one pole per period. Consequently there is one pole in the rational case, one pole modulo the primitive period in the simply periodic case, and one pole on the full elliptic period torus.*

For fixed P of odd degree, there is at most one nonconstant solution with its pole at zero. For even P , the same statement holds after fixing the conserved energy. The only entire solutions are $v = 0$ and $v = -2a_0$.

An even polynomial is one in which every odd power of ρ has coefficient zero, so that $P(D)$ contains only even derivatives. The hypothesis does not cover the mixed operators with p even, and it cannot: Section 10 gives a mixed second-order operator with a meromorphic solution outside $\mathcal{W}$.

3.1 Pole order and the Laurent recurrence

Write a pole expansion at zero as

$$v(z) = \sum_{n \geq 0} c_n z^{n-p}. \quad (3.1)$$

The pole order is forced by dominant balance. A pole of order r makes $P(D)v$ of order $r + p$ and v^2 of order $2r$; the two can cancel only if $r = p$. Comparing the leading coefficients gives

$$c_0 = c_* := -2(-1)^p R_p, \quad R_p = \frac{(2p-1)!}{(p-1)!}. \quad (3.2)$$

For the remaining coefficients, set

$$I_p(n) = \prod_{j=p}^{2p-1} (n-j) + c_*. \quad (3.3)$$

The product is the factor produced by D^p acting on z^{n-p} , and c_* is the factor produced by the cross term of $c_n z^{n-p}$ with the leading pole in $v^2/2$. With $c_n = 0$ for $n < 0$, comparison of the coefficients of z^{n-2p} in (1.1) gives

$$I_p(n)c_n = -\sum_{j=0}^{p-1} a_j (n-2p+j)^{\underline{j}} c_{n-p+j} - \frac{1}{2} \sum_{k=1}^{n-1} c_k c_{n-k}, \quad n \geq 1. \quad (3.4)$$

The underline denotes a falling factorial. The right side contains only coefficients with index less than n , so (3.4) determines c_n whenever $I_p(n) \neq 0$. The positive integer zeros of I_p are the *resonances* of the recurrence: at such an index the recurrence leaves c_n undetermined if its right side vanishes, and has no solution otherwise.

If p is odd, then $c_* = 2R_p$, and there is no positive integer resonance. For integers $1 \leq n < p$, the product in (3.3) is negative with absolute value less than R_p . For $p \leq n \leq 2p-1$ it is zero, and for $n \geq 2p$ it is positive. Hence $I_p(n) \neq 0$ for every positive integer n , and there is exactly one formal Laurent expansion at the marked pole.

Theorem 2 of Eremenko [18] states that an autonomous polynomial differential equation with finitely many possible formal pole expansions and a single monomial of highest total degree has all its meromorphic solutions in $\mathcal{W}$. In (1.1) the only monomial of degree two is $v^2/2$, and for odd p we have just seen that there is exactly one formal pole expansion. Both hypotheses hold at every odd order, for every coefficient vector a .

3.2 The energy argument for an even operator

Let $p = 2m$ and

$$P(D) = D^{2m} + \sum_{j=0}^{m-1} b_j D^{2j}. \quad (3.5)$$

In this subsection b_j denotes the coefficients of the even operator, not those of (2.1). Now $c_* = -2R_p$, and there is exactly one positive integer resonance. Indeed the product in (3.3) is at most R_p for $0 \leq n < p$, is zero for $p \leq n \leq 2p - 1$, and is strictly increasing for $n \geq 2p$; its value at $3p$ is $2R_p = -c_*$. The only positive integer zero of I_p is therefore $n = 3p$, and the possible free Laurent coefficient is

$$h = c_{3p} = [z^{2p}]v.$$

If the recurrence is compatible at $n = 3p$, the equation has a one-parameter family of formal expansions at the pole, one for each value of h , and Eremenko's theorem does not apply to (1.1) itself. The remedy is a first integral whose value determines h .

For $j \geq 1$, define the quadratic differential expression

$$\mathcal{B}_j(v) = \sum_{\ell=0}^{j-2} (-1)^\ell v^{(\ell+1)} v^{(2j-1-\ell)} + \frac{(-1)^{j-1}}{2} (v^{(j)})^2, \quad (3.6)$$

where the sum is empty when $j = 1$. Termwise differentiation gives $D\mathcal{B}_j = v'v^{(2j)}$: every even derivative, multiplied by v' , is an exact derivative. Since P contains only even derivatives, it follows that

$$\mathcal{H}_P(v) = \mathcal{B}_m(v) + \sum_{j=1}^{m-1} b_j \mathcal{B}_j(v) + \frac{b_0}{2} v^2 + \frac{1}{6} v^3 \quad (3.7)$$

satisfies

$$D\mathcal{H}_P = v'E_P(v). \quad (3.8)$$

In particular $\mathcal{H}_P$, the energy, is constant on each solution. It is the energy in the usual sense: an even operator $P(D)$ is formally self-adjoint, (1.1) is then the Euler–Lagrange equation of an autonomous Lagrangian, and $\mathcal{H}_P$ is the conserved quantity that belongs to translation in z . The purely dispersive equations are also exactly those for which (1.1) is invariant under $z \mapsto -z$. The case with no intermediate derivatives is the first integral used by Eremenko et al. [19]. The familiar Kawahara first integral is the case $m = 2$ [14, 48]; the corresponding order-six construction appears in Dang [9].

The energy identity has two consequences for the Laurent expansion. The first is that the resonance is always compatible. To see this, construct (3.1) through index $3p - 1$. The first possible residual of (1.1) is $\Omega_p z^p$. Multiplication by v' gives residue $-pc_*\Omega_p$. By (3.8) this is the residue of a derivative of a Laurent series, and is therefore zero. Hence $\Omega_p = 0$, and h may be chosen arbitrarily.

The second consequence is that the energy determines h . On the full formal solution the constant energy has the form

$$\mathcal{H}_P = \Lambda_p h + \eta_p(P), \quad \Lambda_p = -pc_* I'_p(3p) \neq 0. \quad (3.9)$$

To find the coefficient, linearize the highest-weight part of (3.8) about $c_* z^{-p}$, with perturbation z^{n-p} . If the corresponding energy term is $L(n)z^{n-3p}$, then

$$(n - 3p)L(n) = -pc_* I_p(n).$$

Taking $n \rightarrow 3p$ gives the coefficient in (3.9). Lower derivative terms have different powers of z and cannot contribute to this coefficient. Moreover,

$$I'_p(3p) = 2R_p \sum_{j=p+1}^{2p} \frac{1}{j} > 0.$$

Fixing the energy thus fixes h , and all subsequent Laurent coefficients follow from (3.4). This is the resonance-removal step in Eremenko et al. [19, Lemma 4], with an explicit coefficient for the present operator family; see also Conte et al. [7, Section 3.2] and Conte et al. [8, Section 5].

We can now apply Eremenko's theorem, not to (1.1) but to the level equation $\mathcal{H}_P(v) = H$ with P and H fixed. This equation is autonomous and polynomial, and its unique monomial of highest degree is $v^3/6$. It has exactly one formal pole expansion: differentiating it and dividing by the nonzero Laurent series v' returns $E_P(v) = 0$, so every expansion belongs to the family above, and (3.9) selects the value of h . Every meromorphic solution of (1.1) lies on one energy level and therefore belongs to $\mathcal{W}$.

3.3 The period group and the entire solutions

It remains to show that the poles of one solution form a single orbit under its periods. For either covered case, take two poles z_1, z_2 of the same solution. The functions $v(z + z_1)$ and $v(z + z_2)$ satisfy the same equation and, in the even case, have the same energy. Their Laurent series at zero coincide, so the identity theorem gives

$$v(z + z_1) = v(z + z_2).$$

Every difference of poles is a period. Conversely, every period translates a pole to a pole. The period group is discrete: a sequence of nonzero periods tending to zero would force $v' \equiv 0$. The pole set is therefore one coset of that group. The same argument proves the uniqueness statements of the theorem: two nonconstant solutions with the same operator, the same energy in the even case, and a pole at zero have the same Laurent series there.

Finally, an entire solution in $\mathcal{W}$ must be constant. In the rational case it is a polynomial, and the highest degree of its square cannot be cancelled in (1.1) unless the polynomial is constant. In the rational-exponential case it is a Laurent polynomial in $t = e^{\kappa z}$; the operator $P(\kappa t \partial_t)$ preserves each exponent, whereas squaring doubles an extreme nonzero exponent. An entire elliptic function is constant by compactness. The constants solve $a_0 v + v^2/2 = 0$, which gives $v = 0$ and $v = -2a_0$. This completes the proof of Theorem 3.1.

The same uniqueness shows that, in the two pure subfamilies of Section 2, every wave inherits the reflection symmetry of its equation.

Corollary 3.2 (Waves are symmetric about their poles). *Let v be a nonconstant meromorphic solution of (1.1) and let z_0 be one of its poles.*

- (i) *If P is even, then $v(z_0 + \zeta) = v(z_0 - \zeta)$.*
- (ii) *If p is odd and $P - P(0)$ is odd, then $v(z_0 + \zeta) + P(0) = -(v(z_0 - \zeta) + P(0))$.*

Consequently a solution that is rational in an exponential is a pulse in case (i). In case (ii) its two limits add up to $-2P(0)$, so it is a front unless $P(0) = 0$.

Proof. In case (i) the function $v(z_0 - \zeta)$ solves the same equation, has a pole at $\zeta = 0$, and has the same energy as v , because every term of (3.6) and (3.7) contains an even total number of derivatives. Fixed-energy uniqueness gives (i). In case (ii) the function $-v(z_0 - \zeta) - 2P(0)$ solves the same equation, by Section 2, and has a pole at $\zeta = 0$. For odd p the Laurent expansion at a pole is unique, which gives (ii). The statements about the limits follow by letting ζ tend to the two ends. $\square$

4 Three finite algebraic systems

Theorem 3.1 says that, in the covered cases, a nonconstant meromorphic solution is one of three kinds and has one pole per period. Each kind is a finite-dimensional family of functions: a polynomial in $1/z$, a polynomial in a fixed simply periodic function with one pole, or a constant plus a linear combination of $\wp$ and its derivatives. Substituting the general member of each family into (1.1) therefore gives finitely many polynomial equations in its coefficients. This section writes down the three systems, fixes their normalizations, and proves that they are exactly equivalent to the one-pole solutions (Theorem 4.1). They are instances of the principal-part construction of Demina and Kudryashov [15, 16]; for a fixed operator the same construction reduces to the coefficient comparison of Demina and Kudryashov [11], which we do not need.

The systems all have the same shape. Let V be a polynomial expression in some variables, and let δ be a differentiation rule for those variables, so that the polynomials in them form a differential algebra with derivation δ . Write

$$\mathcal{E}_P(V; \delta) = \delta^p V + \sum_{j=0}^{p-1} a_j \delta^j V + \frac{1}{2} V^2.$$

Each system below is the polynomial identity $\mathcal{E}_P(V; \delta) = 0$ for an appropriate V and δ . A polynomial identity means the vanishing of every coefficient.

4.1 Rational solutions

A rational solution whose only pole is at zero, of order p , and which has no nonconstant polynomial part at infinity is a polynomial of degree p in $T = 1/z$. Since $DT = -T^2$, the equation becomes

$$V_R(T) = b + \sum_{k=1}^p r_k T^k, \quad r_p = c_*, \quad \mathcal{E}_P(V_R; -T^2 \partial_T) = 0. \quad (4.1)$$

The left side of the identity is a polynomial of degree at most $2p$ in T , so this is a finite system in $b, r_1, \dots, r_{p-1}$. No rational solution has a nonconstant polynomial part at infinity: such a part would have its degree doubled by v^2 , while the linear differential expression cannot reach that degree. Consequently (4.1) is the full rational one-pole system.

4.2 Rational functions of an exponential

For $q = \kappa^2 \neq 0$, define

$$T_q(z) = \frac{\kappa}{2} \coth \frac{\kappa z}{2}, \quad DT_q = \frac{q}{4} - T_q^2. \quad (4.2)$$

The expression is independent of the choice of sign of κ . It has period $2\pi i/\kappa$ and one pole per period, at zero, with residue one; it plays the role that $1/z$ plays in the rational case. The finite

system is

$$V_S(T) = \sum_{k=0}^p u_k T^k, \quad u_p = c_*, \quad q \neq 0, \quad \mathcal{E}_P \left(V_S; \left(\frac{q}{4} - T^2 \right) \partial_T \right) = 0. \quad (4.3)$$

To see that every one-pole solution rational in an exponential has this form, write $v = R(t)$, $t = e^{\kappa z}$. The rational function R has finite limits at both $t = 0$ and $t = \infty$. If it had a pole at either end, an extreme Laurent monomial would be doubled by R^2 and could not cancel against $P(\kappa t \partial_t)R$. After a translation, its only finite pole is at $t = 1$, of order p . Such a function is a degree- p polynomial in T_q . The logistic coordinate

$$Q = \frac{1}{1 + e^{-z}}, \quad DQ = Q(1 - Q),$$

is an equivalent coordinate after fixing the rate and translating the pole; $Q(z + i\pi) = T_1(z) + 1/2$. Polynomial logistic representations and their use in constructing travelling waves are developed by Kudryashov [31, 35, 36].

The rational system is the $q = 0$ specialization of (4.3), with $T_0 = 1/z$. We keep $q = 0$ and $q \neq 0$ distinct so that this degeneration is never counted twice.

4.3 Elliptic solutions

Use the Weierstrass algebra

$$Y^2 = 4X^3 - g_2X - g_3, \quad \mathcal{D}X = Y, \quad \mathcal{D}Y = 6X^2 - \frac{g_2}{2}, \quad \Delta = g_2^3 - 27g_3^2 \neq 0. \quad (4.4)$$

Its analytic realization is $X = \wp(z; g_2, g_3)$, $Y = \wp'(z; g_2, g_3)$; see NIST [45, Chapter 23]. Set

$$V_E = b + \sum_{j=0}^{p-2} t_j \mathcal{D}^j X, \quad t_{p-2} = -s_p, \quad s_p = \frac{2(2p-1)!}{((p-1)!)^2}, \quad \mathcal{E}_P(V_E; \mathcal{D}) = 0. \quad (4.5)$$

Using (4.4), every polynomial in X and Y reduces to the normal form $F(X) + YG(X)$. The system consists of the coefficients of F and G set equal to zero; all of them are polynomials with rational coefficients in the displayed unknowns.

The representation is the classical one. An elliptic function with one pole has zero residue there. The functions $1, \wp, \wp', \dots, \wp^{(p-2)}$ have distinct principal parts and span the space of elliptic functions whose only pole is at zero, of order at most p . Subtracting the indicated principal part from a one-pole solution leaves an entire elliptic function, hence a constant. The leading coefficient in (4.5) follows from

$$D^{p-2}\wp = (-1)^{p-2}(p-1)!z^{-p} + O(1).$$

The nonsingularity condition $\Delta \neq 0$ is retained throughout; the singular cubics correspond to the rational and simply periodic systems, which we have already written down.

Theorem 4.1 (Finite matching and exhaustion). *For every $p \geq 2$, the systems (4.1), (4.3), and (4.5) are necessary and sufficient for the respective one-pole solutions of (1.1), with the pole at zero and with the stated normalizations. For the operator classes of Theorem 3.1, these systems, together with the constant solutions, exhaust all globally meromorphic solutions.*

The systems describe exceptional parameter values and continuous families as well as isolated solutions. Their sufficiency follows from substitution, without any genericity assumption.

Proof. Necessity is the principal-part argument given for each kind above. Conversely, a solution of a coefficient system gives a globally meromorphic function by its displayed realization, and the chain rule gives $E_P(v) = 0$. The leading coefficient $c_* \neq 0$ ensures a nonconstant function. In the elliptic case every nonsingular pair (g_2, g_3) has a Weierstrass realization, and the normal form $F(X) + YG(X)$ is unique. The exhaustion statement is Theorem 3.1. $\square$

5 The count of rational-exponential pairs

In this section and the next, P ranges over *all* monic polynomials of degree p , mixed operators included, and we count the pairs (P, v) of Section 2. Nothing in the count requires the hypotheses of Theorem 3.1. Those hypotheses were needed to show that the one-pole solutions are all the meromorphic solutions; the matching equations and their solution counts make sense without them. The pairs that belong to the purely dispersive and purely dissipative subfamilies are counted in Section 5.4.

Recall the normalization: the rate is one, the pole is at $i\pi$ modulo $2\pi i\mathbb{Z}$, and $v \rightarrow 0$ as $e^z \rightarrow 0$. A wave and its mirror image are counted separately when they differ. The functions $Q, DQ, \dots, D^{p-1}Q$ span the degree- p polynomials in Q with zero constant term, since

$$D^j Q = (-1)^j j! Q^{j+1} + \text{lower powers of } Q.$$

The leading pole coefficient (3.2) therefore gives the unique form

$$v_A = s_p A(D)Q, \quad A(\rho) = \rho^d + \alpha_1 \rho^{d-1} + \dots + \alpha_d, \quad d = p - 1. \quad (5.1)$$

The scale s_p was defined in (4.5). This representation is the standard singular-part operator form: $Q = 1/2 + D \log \cosh(z/2)$, so (5.1) is a constant plus a linear differential operator applied to $\log \cosh(z/2)$. Such forms, including the KS case, are discussed by Conte and Musette [4]. We use the coefficients $\alpha_1, \dots, \alpha_d$ of that operator as the coordinates for the enumeration.

The argument runs as follows. The profile v_A solves (1.1) for some monic P exactly when A divides a polynomial built from A by a discrete convolution (Proposition 5.1). The remainder of that division gives d polynomial equations in the d coordinates. When $\ell = 2p - 1$ is prime, these equations become transparent modulo ℓ : they say that A divides $\rho^\ell - \rho$. The monic divisors of degree d of that polynomial are the products $\prod_{s \in S} (\rho - s)$ over the d -element subsets S of $\mathbb{F}_\ell$, and there are $\binom{\ell}{d}$ of them. We show that every solution has coefficients that can be reduced modulo ℓ , so that it reduces to one of these divisors, and that every divisor lifts to exactly one solution (Theorem 5.4). The subset S is the label of the pair. Section 5.3 then reads pulses, fronts and mirror images off the labels, and this gives the count in the two pure subfamilies (Theorem 5.8).

5.1 Discrete convolution and the matching equations

For polynomials F, G , define $F \star G$ by polynomial continuation of

$$(F \star G)(n) = \sum_{j=1}^{n-1} F(j)G(n-j), \quad n \geq 1.$$

The continuation exists by the power-sum formulas: for fixed F, G the right side is a polynomial in n . Write

$$S_A = A \star A.$$

For monic A of degree d ,

$$\deg S_A = 2d + 1, \quad [\rho^{2d+1}]S_A = \beta_d := \frac{(d!)^2}{(2d+1)!}. \quad (5.2)$$

The leading coefficient is the beta integral $\int_0^1 t^d(1-t)^d dt$.

Proposition 5.1 (Polynomial divisibility). *The profile (5.1) solves (1.1) for some monic P if and only if*

$$A \mid S_A. \quad (5.3)$$

The operator is unique and is

$$P_A = \frac{s_p S_A}{2A}. \quad (5.4)$$

Proof. In the half-plane $\operatorname{Re} z < 0$, the logistic function is a geometric series in e^z , and applying $A(D)$ multiplies the n th term by $A(n)$:

$$A(D)Q = \sum_{n \geq 1} (-1)^{n-1} A(n) e^{nz}.$$

Squaring gives the rational-function identity

$$(A(D)Q)^2 = -S_A(D)Q. \quad (5.5)$$

The map $B \mapsto B(D)Q$ is injective: its vanishing would give $B(n) = 0$ for every positive integer n . Consequently (1.1) is equivalent to $2P_A A = s_p S_A$, which has a polynomial solution P_A exactly when A divides S_A . Equation (5.2) and $s_p = 2/\beta_d$ show that P_A is monic of degree p . $\square$

Every P_A has a root among $1, \dots, p$. Let m be the least positive integer with $A(m) \neq 0$; then $m \leq p$, because A has degree $p-1$. As $e^z \rightarrow 0$ the profile begins with $s_p(-1)^{m-1}A(m)e^{mz}$ and its square with e^{2mz} , so the coefficient of e^{mz} in (1.1) gives $P_A(m) = 0$.

The operator determines the profile. This holds for every p , without the hypotheses of Theorem 3.1, because it only uses the leading coefficient of a convolution.

Proposition 5.2 (One profile per operator). *Let A, B be monic polynomials of degree $p-1$ satisfying $A \mid S_A$ and $B \mid S_B$. If $P_A = P_B$, then $A = B$. Consequently a monic operator of degree p admits at most one normalized rational-exponential one-pole profile. Any count of distinct pairs with this normalization is therefore also a count of distinct operators.*

Proof. Suppose $P_A = P_B = P$, and put $h = A - B$. If $h \neq 0$, its degree e is less than $d = p-1$; write $f \neq 0$ for its leading coefficient. By (5.4) and the bilinearity and symmetry of the convolution,

$$S_A - S_B = h \star (A + B) = \frac{2}{s_p} P h.$$

The power-sum calculation in (5.2) gives the leading coefficient $2f \int_0^1 t^e(1-t)^d dt$ on the left. Since P is monic, the right side has leading coefficient $(2/s_p)f = \beta_d f$. Cancelling f would therefore give

$$2 \int_0^1 t^e(1-t)^d dt = \int_0^1 t^d(1-t)^d dt.$$

This is impossible: $e < d$ implies $t^e > t^d$ for $0 < t < 1$, so the left side exceeds twice the right side. Thus $h = 0$. $\square$

Divide S_A by the monic polynomial A :

$$S_A = AC_A + R_A, \quad R_A = \sum_{r=0}^{d-1} R_r(\alpha_1, \dots, \alpha_d) \rho^r. \quad (5.6)$$

The matching equations are $R_0 = \dots = R_{d-1} = 0$: d polynomial equations in the d unknowns $\alpha_1, \dots, \alpha_d$. Their solutions are in bijection with the normalized pairs (P_A, v_A) . We write $\mathcal{X}_p$ for the solution set together with the equations that define it.

Two notions from algebra are used to count. A solution is *simple* if the Jacobian matrix of $(R_0, \dots, R_{d-1})$ with respect to $(\alpha_1, \dots, \alpha_d)$ is invertible there. Every isolated solution has a *multiplicity*. For a square system such as this one it may be pictured as the number of simple solutions into which the point splits under a generic small perturbation of the equations; formally it is the dimension over $\mathbb{C}$ of the local quotient algebra of the equations at that point. It equals one exactly for simple solutions. A *count with multiplicity* is the sum of the multiplicities of the solutions. It equals the number of distinct pairs exactly when every solution is simple.

5.2 The prime-index theorem

Two tools are needed: a grading of the matching equations, and a lemma on reduction modulo a prime.

Weights and the continuous convolution. Give α_j weight j and ρ weight one, so that A is weighted-homogeneous of weight d . This grading reflects how the coefficients scale: replacing the roots of A by λ times themselves multiplies α_j by λ^j . The polynomial S_A has weight at most $2d + 1$, and its part of weight exactly $2d + 1$ is the *continuous* convolution

$$H_A(\rho) = \int_0^\rho A(t)A(\rho - t) dt = \sum_{i,j=0}^d \frac{i!j!}{(i+j+1)!} a_i a_j \rho^{i+j+1}, \quad A(\rho) = \sum_i a_i \rho^i. \quad (5.7)$$

Indeed these are precisely the highest weighted terms of the power sums that define S_A . Monic division preserves the highest weighted part: A is weighted-homogeneous and monic, so each step of long division subtracts a multiple of A of the same weight as the term being removed. Consequently the matching equation R_r has weight at most $2d + 1 - r$, and

$$\text{the part of weight } 2d + 1 - r \text{ of } R_r \text{ is the coefficient of } \rho^r \text{ in } \text{rem}_A H_A. \quad (5.8)$$

In words, the discrete convolution S_A is a lower-order perturbation of the continuous convolution H_A , and far from the origin the matching equations look like the continuous problem $A \mid H_A$. Solutions of the matching equations could escape to infinity only along solutions of that problem. We denote by (R_d) the statement that it has none except the trivial one:

$$A \text{ monic, } \deg A = d, \quad A \mid H_A \implies A(\rho) = \rho^d. \quad (5.9)$$

We call (R_d) *rigidity*. The solution $A = \rho^d$ corresponds to the rational wave $v = c_* z^{-p}$ of the operator $P = D^p$, and rigidity says that (1.1) has no other rational one-pole solution that vanishes at infinity (Section 6, Step 3).

Reduction modulo ℓ . Let $\mathbb{Z}_{(\ell)}$ be the ring of rational numbers whose denominator is prime to ℓ . For a polynomial G in $\alpha_1, \dots, \alpha_d$, write G^{top} for its part of highest weight.

Lemma 5.3 (Reduction modulo ℓ). *Let $G_1, \dots, G_m \in \mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$, and suppose that the reductions modulo ℓ of $G_1^{\text{top}}, \dots, G_m^{\text{top}}$ have no common zero over $\overline{\mathbb{F}}_\ell$ other than the origin. Then:*

- (a) $G_1^{\text{top}}, \dots, G_m^{\text{top}}$ have no common zero over $\mathbb{C}$ other than the origin;
- (b) every common zero of $G_1, \dots, G_m$ with coordinates in an algebraic closure K of $\mathbb{Q}_\ell$ is integral: its coordinates lie in the valuation ring of K .

Proof. Let v be the valuation of K with $v(\ell) = 1$, let $\mathcal{O}$ be its valuation ring, and recall that the residue field of $\mathcal{O}$ is $\overline{\mathbb{F}}_\ell$ and that, K being algebraically closed, every rational number is the valuation of some element of K .

For (b), let $\alpha \in K^d$ be a common zero of the G_i , put $t = \min_j v(\alpha_j)/j$, and suppose that $t < 0$. Choose $\lambda \in K$ with $v(\lambda) = t$, and put $\beta_j = \alpha_j/\lambda^j$. Then every β_j lies in $\mathcal{O}$, and at least one of them is a unit. Decompose $G_i = \sum_{w \leq W_i} G_i^{(w)}$ into its parts of weight w , so that $G_i^{(W_i)} = G_i^{\text{top}}$. A part of weight w satisfies $G_i^{(w)}(\alpha) = \lambda^w G_i^{(w)}(\beta)$. Dividing the equation $G_i(\alpha) = 0$ by λ^{W_i} gives

$$G_i^{\text{top}}(\beta) = - \sum_{w < W_i} \lambda^{w-W_i} G_i^{(w)}(\beta).$$

Each term on the right has positive valuation, because $v(\lambda^{w-W_i}) = (w-W_i)t > 0$ and $G_i^{(w)}(\beta) \in \mathcal{O}$. Reducing modulo the maximal ideal of $\mathcal{O}$, we find that the reduction of β , which is not the origin, is a common zero of the reduced polynomials G_i^{top} . This contradicts the hypothesis. Hence $t \geq 0$, which is (b).

For (a), suppose that the G_i^{top} have a common complex zero other than the origin. They are weighted-homogeneous with rational coefficients, so after fixing one nonzero coordinate to be 1 the weak Nullstellensatz gives such a zero α with coordinates in $\overline{\mathbb{Q}} \subset K$. Define t , λ and β as before, now for either sign of t . Then $G_i^{\text{top}}(\beta) = \lambda^{-W_i} G_i^{\text{top}}(\alpha) = 0$, and the reduction of β is again a forbidden common zero. $\square$

The argument needs the algebraic closure of the residue field; a search for zeros over $\mathbb{F}_\ell$ alone would not suffice.

Theorem 5.4 (Prime-index enumeration). *Let $\ell = 2p - 1$ be prime, and put $d = p - 1$.*

- (a) *The matching equations have exactly $\binom{\ell}{d} = N_p$ solutions over $\mathbb{C}$, and every solution is simple. Thus there are exactly N_p rational-exponential pairs, and they are distinct.*
- (b) *The coefficients of every solution are algebraic numbers that are integral at ℓ . After a choice of an ℓ -adic embedding of the algebraic numbers, reduction modulo ℓ is a bijection between the solutions and the d -element subsets $S \subset \mathbb{F}_\ell$:*

$$\overline{A}(\rho) = \prod_{s \in S} (\rho - s). \tag{5.10}$$

- (c) *Rigidity (R_d) holds.*

We call S the *label* of the pair.

Proof. We work over $\mathbb{Z}_{(\ell)}$. The proof has four steps: a congruence for S_A modulo ℓ ; the solution of the reduced equations over $\mathbb{F}_\ell$; integrality, which shows that every solution reduces to one of them; and lifting, which shows that each of them comes from exactly one solution.

Step 1: the congruence. Induction using $(D - n)Q^n = -nQ^{n+1}$ gives

$$Q^n = \frac{(-1)^{n-1}}{(n-1)!} \prod_{j=1}^{n-1} (D - j)Q.$$

Write $(A(D)Q)^2 = \sum_{n=2}^{2p} b_n Q^n$. Each b_n is an integer polynomial in the coefficients of A , and $b_{2p} = (d!)^2$. By (5.5),

$$S_A(\rho) = \sum_{n=2}^{2p} \frac{(-1)^n b_n}{(n-1)!} \prod_{j=1}^{n-1} (\rho - j).$$

Only the last denominator, $(2p-1)! = \ell!$, contains ℓ . Consequently ℓS_A has coefficients in $\mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$, and

$$\ell S_A(\rho) \equiv \frac{(d!)^2}{(\ell-1)!} \prod_{j=1}^{\ell} (\rho - j) \equiv (-1)^d (\rho^\ell - \rho) \pmod{\ell}. \quad (5.11)$$

The scalar follows by pairing j with $\ell - j$ in $(\ell-1)!$, and the product is $\rho^\ell - \rho$ because it runs over all residues modulo ℓ . The right side does not depend on A : this is what makes the prime index special.

Step 2: the reduced equations. Monic division preserves integrality, so the scaled remainder equations $\ell R_r = 0$ have coefficients in $\mathbb{Z}_{(\ell)}$, and by (5.11) they reduce exactly to $\bar{A} \mid \rho^\ell - \rho$. Since $\rho^\ell - \rho = \prod_{s \in \mathbb{F}_\ell} (\rho - s)$, its monic divisors of degree d are the $\binom{\ell}{d}$ polynomials (5.10); all of them are defined over $\mathbb{F}_\ell$, and they exhaust the solutions over its algebraic closure. Each of them is a simple solution of the reduced equations. Write $\rho^\ell - \rho = \bar{A}C$. For a variation h with $\deg h < d$, the derivative of the remainder map is, up to the nonzero scalar $(-1)^d$,

$$h \longmapsto -\text{rem}_{\bar{A}}(Ch), \quad (5.12)$$

because the dividend (5.11) does not vary with A . Since $\rho^\ell - \rho$ is squarefree, $\bar{A}$ and C are coprime, so multiplication by C is invertible modulo $\bar{A}$, and the map (5.12) is invertible.

Step 3: integrality and rigidity. We apply Lemma 5.3 to $G_r = \ell R_r$, $0 \leq r < d$. By (5.8), G_r^{top} is the coefficient of ρ^r in $\text{rem}_A(\ell H_A)$. Taking the part of weight ℓ in (5.11) gives

$$\ell H_A(\rho) \equiv (-1)^d \rho^\ell \pmod{\ell}, \quad (5.13)$$

so the reduced polynomials G_r^{top} vanish together exactly when $\bar{A} \mid \rho^\ell$, that is, when $\bar{A} = \rho^d$, which is the origin $\alpha_1 = \dots = \alpha_d = 0$. The hypothesis of the lemma holds. Part (a) of the lemma says that $\text{rem}_A H_A = 0$ only for $A = \rho^d$, which is (R_d) . Part (b) says that every solution of the matching equations with coordinates in K is integral. Such a solution therefore has a reduction modulo ℓ , and by Step 2 the reduction is one of the polynomials (5.10).

Step 4: lifting. Let $L \subset K$ be a finite extension of $\mathbb{Q}_\ell$, with ring of integers $\mathcal{O}_L$. The system $(\ell R_r)_{0 \leq r < d}$ has coefficients in $\mathbb{Z}_{(\ell)} \subset \mathbb{Z}_\ell$, and by Step 2 its Jacobian determinant is a unit at each of the $\binom{\ell}{d}$ reduced solutions. The multivariable Hensel lemma [3], whose proof by Newton iteration applies in any complete discrete valuation ring, then gives exactly one solution in $\mathcal{O}_L^d$.

above each reduced solution, and this solution already lies in $\mathbb{Z}_\ell^d$. Every solution in K^d has its coordinates in some such L , and lies in $\mathcal{O}_L^d$ by Step 3. Hence reduction modulo ℓ is a bijection between the solutions in K^d and the reduced solutions: there are exactly $\binom{\ell}{d}$ of them, and the Jacobian determinant is nonzero at each.

The matching equations have rational coefficients, and we have shown that they have finitely many solutions over the algebraically closed field K . Their solutions over any algebraically closed field of characteristic zero are then the same finite set of points with coordinates in $\overline{\mathbb{Q}}$. In particular they have exactly $\binom{\ell}{d}$ solutions over $\mathbb{C}$, all simple. This proves (a) and (b); part (c) was proved in Step 3. $\square$

Remark 5.5 (Formal verification). Parts (a) and (c) of Theorem 5.4 have been verified in Lean 4 against the Mathlib library, both for the matching equations and for the pairs (P, v) themselves; Appendix A describes the files and the scope of the verification. The formal proof follows the route above: the congruence, integrality of every complex solution, injectivity of the reduction, and Hensel lifting of every label. For rigidity it normalizes the roots of A instead of its coefficients, which avoids the auxiliary element λ .

The labels (5.10) describe reductions of algebraic coefficients; they are not assertions that the complex roots of A , or the characteristic roots of P_A , are integers. For example, $A = \rho^2 + 1/11$ at $p = 3$ reduces to $(\rho - 2)(\rho - 3)$ modulo five, so its label is $\{2, 3\}$.

5.3 Reading the labels: pulses, fronts and mirror images

The endpoints of v_A are its limits as $z \rightarrow \mp\infty$, namely 0 and $s_p A(0)$. Pairs with $A(0) = 0$ have equal endpoints and are pulses; the others are fronts. The pulses are needed separately, because the elliptic pairs of Section 6 degenerate onto them.

Proposition 5.6 (Pulses and fronts). *Let $\ell = 2p - 1$ be prime. A pair is a pulse if and only if its label contains 0. Consequently exactly*

$$S_p := \binom{2p-2}{p-2}$$

of the N_p pairs are pulses, and the remaining $\binom{2p-2}{p-1}$ are fronts. Every pulse is a simple solution of the pulse equations (5.15) below, and $P_A(0) \neq 0$ at every pulse. These are counts over $\mathbb{C}$.

Proof. If $A(0) = 0$, then $\overline{A}(0) = 0$, so 0 belongs to the label. For the converse we show that every label containing 0 is the label of a pulse; since a label determines its pair, this pulse is the pair with that label.

Polynomial summation gives $S_A(0) = -A(0)^2$, so the constant remainder satisfies the polynomial identity

$$R_0 = -A(0)(A(0) + C_A(0)). \quad (5.14)$$

On the hyperplane $\alpha_d = A(0) = 0$ the equation $R_0 = 0$ therefore holds identically, and the pulses are the solutions of the *pulse equations*

$$R_1 = \cdots = R_{d-1} = 0 \quad \text{in the unknowns } \alpha_1, \dots, \alpha_{d-1}, \quad \alpha_d = 0. \quad (5.15)$$

We write $\mathcal{Z}_p$ for this system together with its solution set. Let S be a label containing 0, and $\overline{A}$ the corresponding divisor of $\rho^\ell - \rho$. By (5.14), at the reduced solution $\overline{A}$ the row of the

Jacobian matrix that belongs to R_0 has a single nonzero entry, $\pm C(0)$ in the column of α_d ; here $C = (\rho^\ell - \rho)/\bar{A}$, and $C(0) \neq 0$ because $\rho^\ell - \rho$ is squarefree. The full Jacobian is invertible by Step 2 of the proof of Theorem 5.4, so the minor obtained by deleting that row and that column is invertible as well. This minor is the Jacobian of the pulse equations. Hensel's lemma therefore lifts $\bar{A}$ to a solution of the pulse equations over $\mathbb{Z}_\ell$. Completed by $\alpha_d = 0$, it is a solution of the full matching system above $\bar{A}$, hence, by the uniqueness in Step 4, the solution with label S . That solution is thus a pulse, and the Jacobian determinant of the pulse equations is a unit there.

The number of d -element subsets of $\mathbb{F}_\ell$ containing 0 is $\binom{\ell-1}{d-1} = S_p$. Finally ℓC_A reduces to $(-1)^d C$, so $C_A(0) \neq 0$ at every pulse, and $P_A(0) = s_p C_A(0)/2 \neq 0$. $\square$

5.4 Mirror images: the two pure subfamilies

The numbers N_p and M_p refer to the full monic family. We now count the pairs whose equation is purely dispersive or purely dissipative. These are the pairs that coincide with their own mirror image under the reflection $z \mapsto -z$. On the labels the reflection is $S \mapsto -S$, and the number of labels with $S = -S$ is again a binomial coefficient.

For a monic polynomial A of degree $d = p - 1$ put

$$(\iota A)(\rho) = (-1)^d A(-\rho), \quad \text{that is,} \quad \alpha_j \mapsto (-1)^j \alpha_j. \quad (5.16)$$

Proposition 5.7 shows that ι is the reflection $z \mapsto -z$ written in the coordinates α_j . It reads the wave v_A in the opposite direction and then restores the normalization: it moves the zero limit back to the end $e^z \rightarrow 0$, and for odd p it changes the sign of the profile.

Proposition 5.7 (Reflection). *For every monic A of degree d ,*

$$S_{\iota A}(\rho) = -S_A(-\rho) - 2A(0)A(-\rho), \quad R_{\iota A}(\rho) = -R_A(-\rho). \quad (5.17)$$

Consequently ι maps solutions of the matching equations to solutions, and simple solutions to simple solutions. At a point of $\mathcal{X}_p$,

$$v_{\iota A}(z) = (-1)^p (v_A(-z) - s_p A(0)), \quad P_{\iota A}(\rho) = (-1)^p (P_A(-\rho) + s_p A(0)). \quad (5.18)$$

The pairs A and ιA are the mirror images that N_p counts separately. For $p = 3$ the involution is $(a, b) \mapsto (-a, b)$ in Table 2.

Proof. Since $Q(-z) = 1 - Q(z)$, and D changes sign under $z \mapsto -z$,

$$(A(D)Q)(-z) = A(0) - A(-D)Q = A(0) - (-1)^d (\iota A)(D)Q.$$

This is the first formula in (5.18). Evaluate (5.5) at $-z$, and use (5.5) for ιA on the left:

$$A(0)^2 - 2A(0)A(-D)Q - S_{\iota A}(D)Q = -S_A(0) + S_A(-D)Q.$$

Every $B(D)Q$ is a polynomial in Q without constant term, so letting $z \rightarrow -\infty$ gives $S_A(0) = -A(0)^2$. The remaining terms, and the injectivity of $B \mapsto B(D)Q$, give the first identity in (5.17). Now substitute $S_A = AC_A + R_A$ at $-\rho$:

$$S_{\iota A}(\rho) = (\iota A)(\rho) (-1)^p (C_A(-\rho) + 2A(0)) - R_A(-\rho).$$

Division by the monic polynomial ιA is unique. Hence

$$R_{\iota A}(\rho) = -R_A(-\rho), \quad C_{\iota A}(\rho) = (-1)^p(C_A(-\rho) + 2A(0)).$$

The first relation reads $R_r(\iota\alpha) = (-1)^{r+1}R_r(\alpha)$, so the linear map (5.16) preserves the ideal of the matching equations. The second relation and (5.4) give the formula for $P_{\iota A}$. $\square$

A pair is fixed by ι when $\alpha_j = 0$ for every odd j . We write $\mathcal{X}_p^{\text{sym}}$ for the set of these pairs, and put

$$F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}. \quad (5.19)$$

Its first values are 1, 2, 3, 6, 10, 20 for $p = 2, \dots, 7$; Table 1 lists the equations to which they belong.

Theorem 5.8 (Count in the two pure subfamilies). (i) *For every p , a pair A coincides with its mirror image, $\iota A = A$, if and only if P_A is reflection-symmetric. Thus $\mathcal{X}_p^{\text{sym}}$ consists of the rational-exponential pairs of the purely dispersive equations when p is even, and of the purely dissipative equations when p is odd.*

(ii) *Let $\ell = 2p - 1$ be prime. If A has label S , then ιA has label $-S$. Hence $\iota A = A$ if and only if $S = -S$, and there are exactly F_p such pairs. Each of them is a simple solution of the matching equations of its subfamily.*

(iii) *For even p , every pair of $\mathcal{X}_p^{\text{sym}}$ is a pulse. For odd p with $2p - 1$ prime, every pair of $\mathcal{X}_p^{\text{sym}}$ is a front; for every odd p , every pair of $\mathcal{X}_p^{\text{sym}}$ with real coefficients is a front.*

In plain terms, the pairs that coincide with their mirror image are exactly the pairs of the purely dispersive equations, for even p , or of the purely dissipative equations, for odd p . When $2p - 1$ is prime there are exactly F_p of them, and they are the pairs whose label is symmetric. The rational-exponential waves of a purely dispersive equation are pulses, for every even p . Those of a purely dissipative equation are fronts: when $2p - 1$ is prime, and for waves with real coefficients at every odd p . Elliptic waves occur in both subfamilies; the distinction between pulses and fronts does not apply to them. Section 7 records what is known at the remaining orders.

Proof. (i). If $\iota A = A$, then (5.18) gives $P_A(\rho) = (-1)^p(P_A(-\rho) + s_p A(0))$. For even p the polynomial A is odd, so $A(0) = 0$ and P_A is even. For odd p the sum $P_A(\rho) + P_A(-\rho)$ is the constant $-s_p A(0)$. In both cases P_A is reflection-symmetric, and

$$s_p A(0) = -2P_A(0) \quad (p \text{ odd}). \quad (5.20)$$

Conversely, let $P = P_A$ be reflection-symmetric, and apply Corollary 3.2 at the pole $i\pi$, using the period $2\pi i$. If p is even, then $v_A(-z) = v_A(z)$. Letting $z \rightarrow -\infty$ gives $A(0) = 0$, and (5.18) becomes $v_{\iota A} = v_A$. If p is odd, then $-v_A(-z) = v_A(z) + 2P(0)$, and (5.18) becomes $v_{\iota A}(z) = v_A(z) + 2P(0) + s_p A(0)$. Letting $z \rightarrow -\infty$ shows that the constant vanishes. In both cases $v_{\iota A} = v_A$, hence $\iota A = A$ by the uniqueness of the form (5.1).

(ii). The map (5.16) negates the roots of $\bar{A}$, so the label of ιA is $-S$. A label determines its pair (Theorem 5.4), so $\iota A = A$ exactly when $S = -S$. To count these labels, note that the nonzero residues modulo $\ell = 2d + 1$ form d pairs $\{s, -s\}$. A set $S = -S$ of d elements consists of $d/2$ such pairs if d is even, and of 0 and $(d - 1)/2$ pairs if d is odd. In both cases there are $\binom{d}{\lfloor d/2 \rfloor} = F_p$ sets.

On the subspace $\{\alpha_j = 0, j \text{ odd}\}$ we have $\iota A = A$, and (5.17) shows that R_A is an odd polynomial. The equations $R_r = 0$ with r even therefore hold identically there, and the matching equations of the subfamily are the equations $R_r = 0$, r odd, in the unknowns α_j , j even. Differentiating the relation $R_r(\iota\alpha) = (-1)^{r+1}R_r(\alpha)$ at a fixed point shows that $\partial R_r/\partial \alpha_j = 0$ unless $j \equiv r + 1$ modulo two. The Jacobian matrix of the full system therefore splits into two square blocks. It is invertible, so both blocks are, and one of them is the Jacobian of the equations of the subfamily.

(iii). For even p the polynomial A is odd, so $A(0) = 0$. Let p be odd. If $2p - 1$ is prime, a label with $S = -S$ has an even number d of elements, so it does not contain 0, and the pair is a front by Proposition 5.6. For a general odd p , suppose that a point of $\mathcal{X}_p^{\text{sym}}$ with real coefficients is a pulse, $A(0) = 0$. Then $P_A(0) = 0$ by (5.20). Now v_A is real on the real axis, and it tends to zero with all its derivatives at both ends. Integrating (1.1) along the real axis, and using $P_A(0) = 0$, gives $\int v_A^2 dz = 0$. Hence $v_A = 0$, which is impossible. $\square$

6 The count of elliptic pairs

We now count the elliptic pairs of Section 2. Fix a lattice $\Lambda \subset \mathbb{C}$, including its scale, and put $g_i = g_i(\Lambda)$. We place the pole at zero, allow P to vary over all monic polynomials of degree p , and retain the constant term of the profile. Neither reflection nor a lattice automorphism is factored out. By Section 4, every profile has the form

$$V = -s_p \mathcal{D}^{p-2} X + \sum_{j=0}^{p-3} u_j \mathcal{D}^j X + b. \quad (6.1)$$

Its only pole on $\mathbb{C}/\Lambda$ is zero, of order p . In particular Λ is its full period lattice. Let $\mathcal{E}_p(g_2, g_3)$ denote the matching equations obtained by substituting (6.1) in (1.1), in the unknowns $a_0, \dots, a_{p-1}$, $u_0, \dots, u_{p-3}$ and b , together with their solution set. The operator coefficients may equivalently be eliminated first, as in the proof below. Multiplicity and simplicity are understood as in Section 5.

Theorem 6.1 (Fixed-lattice elliptic enumeration). *Let $2p - 1$ be prime.*

- (a) *On every nonsingular lattice the elliptic pairs are finite in number, and counted with multiplicity there are*

$$M_p := 2 \binom{2p-2}{p-2} = 2S_p \quad (6.2)$$

of them.

- (b) *For (g_2, g_3) in a nonempty Zariski-open subset of $\{g_2^3 - 27g_3^2 \neq 0\}$, all of them are simple. Thus a generic lattice admits exactly M_p distinct elliptic pairs.*

The number M_p is twice the number S_p of pulses of Proposition 5.6, and the proof shows why: when one period of the lattice becomes infinite, every elliptic pair becomes a pulse on one of its two constant backgrounds, and every pulse on either background arises in this way exactly once. On an arbitrary lattice the assertion is a count with multiplicities. Exceptional lattices, on which distinct pairs merge, do occur and cannot be discarded when counting distinct solutions; Section 9.2 determines them at order three.

The proof has five steps. We eliminate the operator, which leaves $p - 1$ equations in the $p - 1$ profile unknowns, with coefficients that are polynomials in g_2, g_3 . We assign weights to

the unknowns, under which the highest weighted parts of the equations are obtained by setting $g_2 = g_3 = 0$. At this cusp both periods are infinite and the problem is the rational one, which has only the trivial solution by rigidity; this means that on no lattice can solutions escape to infinity. It follows that the pairs form a finite flat family over the whole (g_2, g_3) -plane, singular cubics included, so that the number of pairs counted with multiplicity is the same on every fibre. Finally we count on the nodal cubic, where the elliptic problem becomes the pulse case of the rational-exponential problem, with two choices of background; its solutions are simple by Proposition 5.6, and simplicity spreads from the nodal parameter value to nearby nonsingular parameters.

Proof. Step 1: eliminating the operator. The coefficient at pole order $2p$ vanishes by the choice of s_p . The coefficients at orders $2p-1, \dots, p$ determine $a_{p-1}, \dots, a_0$ successively. At each step the coefficient of the new a_j is the nonzero rational leading coefficient of $D^j V$. This is a polynomial elimination, valid over any $\mathbb{Q}$ -algebra. The residual has pole order at most $p-1$ and zero residue. It lies in the span of

$$\{1\} \cup \{\mathcal{D}^j X : 0 \leq j \leq p-3\}.$$

Hence $p-1$ equations remain in the $p-1$ profile coefficients u_j, b .

Step 2: weights. Give D, X, g_2, g_3 weights $1, 2, 4, 6$, respectively, and give the profile weight p . Then u_j has weight $p-j-2$, and b has weight p . After the triangular elimination, the residual principal-part coefficients and the residual Laurent constant have weights

$$\begin{array}{c|c} \text{variables} & 1, 2, \dots, p-2, p \\ \text{equations} & p+1, p+2, \dots, 2p-2, 2p. \end{array} \quad (6.3)$$

The Laurent expansions and the eliminations are polynomial in g_2, g_3 , so these statements extend to the singular cubics. For fixed g_2, g_3 , the highest weighted parts of the equations in the profile variables are obtained by setting $g_2 = g_3 = 0$, since g_2 and g_3 carry positive weight.

Step 3: the cusp. At the cusp $g_2 = g_3 = 0$, $X = z^{-2}$ and $Y = -2z^{-3}$, so the profile is a rational function. Write $V = b + f(z)$, where f has its sole pole at zero and vanishes at infinity. The constant and decaying parts of the equation give

$$a_0 b + \frac{1}{2} b^2 = 0, \quad (P+b)(D)f + \frac{1}{2} f^2 = 0.$$

The leading term of f at infinity forces $a_0 + b = 0$. It follows that $b = a_0 = 0$. Under the polynomial Laplace correspondence

$$\sum_k A_k \rho^k \mapsto \sum_k k! A_k z^{-k-1},$$

differentiation corresponds to multiplication by $-\rho$, and multiplication of the rational functions corresponds to continuous convolution. After a nonzero scalar normalization, the equation for f therefore gives $A \mid H_A$ for a monic A of degree $p-1$. By rigidity, which holds by Theorem 5.4(c), $A = \rho^{p-1}$. Thus $f = c_* z^{-p}$, and all the variables in (6.1) vanish. The leading equations have no nonzero common zero.

Step 4: a finite flat family. Let $E_1, \dots, E_{p-1}$ be the residual equations of Step 1. They are polynomials in the $p-1$ profile unknowns and in g_2, g_3 , weighted-homogeneous for the weights of Step 2, and

$$\mathcal{A} = \mathbb{C}[g_2, g_3][u_0, \dots, u_{p-3}, b] / (E_1, \dots, E_{p-1})$$

is the coordinate ring of the family of all the systems $\mathcal{E}_p(g_2, g_3)$. By Step 3 the cusp system has the origin as its only solution, so $\mathcal{A}/(g_2, g_3)\mathcal{A}$ is a finite-dimensional vector space. The graded form of Nakayama's lemma then shows that $\mathcal{A}$ is a finitely generated module over $\mathbb{C}[g_2, g_3]$: homogeneous elements whose classes span the cusp ring generate it. In particular every fibre is finite, which proves the finiteness in (a) and makes precise the statement that solutions do not escape to infinity on any lattice.

The family is also flat. It is defined by $p - 1$ equations in $p + 1$ variables, so each of its components has dimension at least two, and it is finite over the plane, so its dimension is exactly two. Hence $E_1, \dots, E_{p-1}$ is a regular sequence and $\mathcal{A}$ is Cohen–Macaulay. A Cohen–Macaulay algebra that is finite over a regular ring of the same dimension is flat over it [41, Theorem 23.1], and a finite flat module over $\mathbb{C}[g_2, g_3]$ is locally free. Consequently the number of solutions of $\mathcal{E}_p(g_2, g_3)$, counted with multiplicity, is the same at every point of the (g_2, g_3) -plane, the singular cubics included. At the cusp itself all of this multiplicity sits at the single solution where every unknown vanishes.

Step 5: the node. To find this common number, and to prove generic simplicity, consider the nodal parameters

$$g_2 = \frac{1}{12}, \quad g_3 = -\frac{1}{216}. \quad (6.4)$$

In the logistic coordinate they are realized by

$$X = Q^2 - Q + \frac{1}{12}, \quad Y = (2Q - 1)Q(1 - Q), \quad DQ = Q(1 - Q).$$

The nodal coordinate ring is

$$\mathbb{C}[X, Y] = \{f \in \mathbb{C}[Q] : f(0) = f(1)\}.$$

Indeed the right side is $\mathbb{C} + Q(1 - Q)\mathbb{C}[Q]$; it is generated by X, Y , since, with $S = Q(1 - Q)$, $S = 1/12 - X$, $SQ = (Y + S)/2$, and $Q^2 = Q - S$. The pole filtration is the ordinary polynomial degree in Q . Consequently the nodal specialization of (6.1) has the unique form

$$V = \beta + v_A, \quad A \text{ monic of degree } p - 1, \quad A(0) = 0,$$

where β is the common endpoint. The coefficient change is invertible and affine over $\mathbb{Q}$.

The matching equations become, as systems of equations with their multiplicities,

$$A \in \mathcal{Z}_p, \quad P = P_A - \beta, \quad \beta(\beta - 2P_A(0)) = 0. \quad (6.5)$$

To see this directly, the constant equation is $P(0)\beta + \beta^2/2 = 0$. The remaining polynomial identity is $(P + \beta)(D)v_A + v_A^2/2 = 0$, equivalent by Proposition 5.1 to $A \mid S_A$ and $P + \beta = P_A$. The coefficient changes and monic division used here work over arbitrary $\mathbb{C}$ -algebras, so the correspondence preserves local multiplicities.

By Proposition 5.6, $\mathcal{Z}_p$ consists of S_p simple points, with $P_A(0) \neq 0$ at every point. Each quadratic in (6.5) then has two distinct roots, and the nodal fibre consists of $2S_p$ simple points. By Step 4 every fibre has the same number of points counted with multiplicity, namely $2S_p = M_p$. This proves (a).

In the finite coefficient family, the image of the closed locus where the matching Jacobian vanishes is closed. Its complement contains (6.4), and hence meets the nonsingular-lattice locus in a nonempty open set. Every fibre in that open set consists of simple points, which proves (b). $\square$

The factor two in M_p retains the two choices of background in (2.5). That transformation preserves the evolution equation and the lattice; it changes the operator P and the speed. At the nodal fibre it exchanges the two roots in (6.5), which are distinct. The two backgrounds are therefore distinct on a generic lattice, and the M_p pairs represent $M_p/2$ elliptic waves, up to translations and Galilean shifts, each counted with both of its backgrounds. On special nonsingular lattices the two backgrounds may coalesce, as Section 9.2 shows.

6.1 Elliptic pairs of purely dispersive equations

Corollary 6.2. *Let p be even, with $2p - 1$ prime. On every nonsingular lattice, the elliptic pairs whose operator is an even polynomial are those with $u_j = 0$ for odd j in (6.1), and counted with multiplicity there are*

$$2F_p \tag{6.6}$$

of them. On a generic lattice they are distinct.

Proof. An even operator has an even profile, by Corollary 3.2. Conversely, if the profile (6.1) is even, the odd part of (1.1) is $\sum_{j \text{ odd}} a_j D^j V = 0$; the functions $D^j V$ have distinct pole orders, so $a_j = 0$ for odd j . On the subspace $u_j = 0$, j odd, the residual of Step 1 is even, and there remain the equations and unknowns of even weight in (6.3):

$$\begin{array}{l|l} \text{variables} & 2, 4, \dots, p-2, p \\ \text{equations} & p+2, p+4, \dots, 2p-2, 2p. \end{array}$$

There are $p/2$ of each. The omitted leading forms vanish identically on the subspace, so the remaining ones have only the trivial common zero, by Step 3, and Step 4 applies without change: the even pairs form a finite flat family over the (g_2, g_3) -plane. At the node, reflection is $Q \mapsto 1 - Q$. An even nodal profile is $\beta + v_A$ with $\iota A = A$, so the nodal fibre is (6.5) with $A \in \mathcal{X}_p^{\text{sym}}$. By Theorem 5.8 there are F_p such A , all pulses, each a simple solution of the equations of the subfamily, and $P_A(0) \neq 0$ by Proposition 5.6. The nodal fibre thus has $2F_p$ simple points, and Step 5 applies without change. $\square$

For odd p there is no fixed-lattice analogue. A purely dissipative equation has a profile that is $-P(0)$ plus a function odd about the pole, and its matching system has one equation more than it has unknowns. An exact computation on a sample lattice shows that for $p = 3, 5, 7$ it has no solution there, and hence none on a generic lattice. At $p = 3$ such pairs occur exactly when $g_2 = 0$, as (9.7) shows for $B = 0$.

7 Other orders

The proofs of Sections 5 and 6 use the prime $\ell = 2p - 1$ twice: to show that solutions cannot escape to infinity, which is rigidity, and to show that every pair is simple. This section records what is known when $2p - 1$ is not prime. The proofs use a weighted form of Bézout's theorem instead of the labels; they are given in the companion note [anc/general_orders.pdf](#), which is part of the source package and contains the general-order versions of Proposition 5.6, Theorem 5.8, Theorem 6.1 and Corollary 6.2.

Proposition 7.1 (Every order). *Let $p \geq 2$.*

- (i) *The isolated solutions of the matching equations, counted with multiplicity, number at most N_p , and those with $\iota A = A$ at most F_p .*
- (ii) *If rigidity (R_{p-1}) holds, all the systems below have finitely many solutions, and counted with multiplicity there are exactly N_p rational-exponential pairs, S_p solutions of the pulse equations (5.15), F_p solutions of the matching equations of the pure subfamily, M_p elliptic pairs on every nonsingular lattice and, for even p , $2F_p$ elliptic pairs with an even operator.*
- (iii) *If, in addition, every rational-exponential pair is simple, then all the conclusions of Theorem 5.4(a), Proposition 5.6, Theorem 5.8, Theorem 6.1 and Corollary 6.2 hold at this p , except for the labelling.*

The weights behind (i) are those of Section 5.2: the equations $R_0, \dots, R_{d-1}$ have weights $2d+1, \dots, d+2$ in unknowns of weights $1, \dots, d$, and the quotient of the two products is $\binom{2d+1}{d}$.

Two hypotheses thus remain at a general order, and this is what is known about them.

- *Rigidity.* (R_d) is proved when $2d+1$ is a prime or a prime power. Exact integer certificates in the source package establish it for every $d \leq 72$, and finite-field Gröbner computations, checked by independent pipelines, for every $d \leq 572$ (repository, directory `certificates/rigidity_572/`). Hence the counts with multiplicity of Proposition 7.1(ii) hold for every $2 \leq p \leq 573$.
- *Simplicity.* At the composite indices below 25, exact computations show that all pairs are simple: at $p=5$ by a Gröbner basis over $\mathbb{Q}$, at $p=8$ over a finite field with Hensel lifting, and at $p=11$ by certificates over the 19-adic numbers (Appendix A). Inside the two pure subfamilies the computation is much smaller, and it shows that the F_p pairs are distinct for every $p \leq 14$.

Corollary 7.2 (Orders up to twelve). *For every $2 \leq p \leq 12$ there are exactly N_p rational-exponential pairs, all simple, of which S_p are pulses and F_p belong to the pure subfamily of that order; and a generic lattice carries exactly M_p distinct elliptic pairs. In particular this holds for all the equations of Table 1.*

The first order at which distinctness is not known is $p=13$. We conjecture that rigidity and simplicity hold in every order (Conjecture 10.1).

8 What the count means for a single equation

The count lets the operator vary. We now read it for one evolution equation (2.1) with fixed complex coefficients. Recall that (2.1) is invariant under translations and under the Galilean shifts $u \mapsto u+k$, $x \mapsto x-kt$.

Proposition 8.1 (The waves of a fixed equation). *Let $\lambda \neq 0$. Up to a translation and a Galilean shift, the travelling waves of (2.1) that are rational in $e^{\lambda(x-ct)}$, with one pole modulo $(2\pi i/\lambda)\mathbb{Z}$, are*

$$u = b_p \lambda^p v_A(\lambda(x-ct)), \quad c = -b_p \lambda^p P_A(0), \quad (8.1)$$

where A ranges over the points of $\mathcal{X}_p$ that satisfy

$$\frac{b_j}{b_p} = \lambda^{p-j} [\rho^j] P_A, \quad 1 \leq j \leq p-1. \quad (8.2)$$

The data (A, λ) and $(\iota A, -\lambda)$ describe the same wave.

Proof. Such a wave has a finite limit as $e^{\lambda(x-ct)} \rightarrow 0$, by the argument of Section 4, and a Galilean shift makes this limit zero. Then $w_* = 0$ in (2.3), and (2.4) becomes (8.2) together with the formula for c . After a translation v has the normalization of Section 2, so $v = v_A$ and $P = P_A$ by Proposition 5.1. The last assertion follows from (5.18): the two waves differ by the constant $b_p \lambda^p s_p A(0)$, which is a Galilean shift. $\square$

Let $\pi(A) = ([\rho^{p-1}]P_A, \dots, [\rho^1]P_A)$ be the vector of lower coefficients of P_A , and let $\beta = (b_{p-1}/b_p, \dots, b_1/b_p)$ be the coefficient vector of the equation; this β is not the constant β of Section 6. Condition (8.2) says that β is obtained from $\pi(A)$ by multiplying its k th entry by λ^k , for some $\lambda \neq 0$. As λ varies these vectors trace a curve, the orbit $\mathbb{C}^* \pi(A)$, and rescaling x moves an equation along it. This is the reason for fixing the rate: without it every pair would be a one-parameter family. Each of the N_p pairs marks one orbit, and several pairs may mark the same orbit.

Corollary 8.2 (Solvable equations). *Assume that the matching equations have finitely many solutions. This holds when $2p - 1$ is prime, and for every $p \leq 573$ (Section 7).*

- (i) *Equation (2.1) has a travelling wave that is rational in an exponential, with one pole per period, if and only if β lies in one of the orbits $\mathbb{C}^* \pi(A)$, $A \in \mathcal{X}_p$. There are at most $(N_p + F_p)/2$ such orbits.*
- (ii) *If $\beta \neq 0$, there are finitely many such waves up to translations and Galilean shifts.*
- (iii) *If $p \geq 3$, the orbits form a set of dimension at most one in $\mathbb{C}^{p-1}$. A generic equation (2.1) has no such wave.*

Proof. Part (i) restates Proposition 8.1. For the bound, $\pi(\iota A) = (-1) \cdot \pi(A)$ by (5.18), so A and its mirror image ιA give the same orbit. The number of classes $\{A, \iota A\}$ is half the number of points plus half the number of fixed points, and these two numbers are at most N_p and F_p , by Proposition 7.1(i). For (ii), if $\beta \neq 0$ and $\beta \in \mathbb{C}^* \pi(A)$, some coordinate $[\rho^j]P_A$ is nonzero, and (8.2) leaves at most $p - j$ values of λ . Part (iii) is a dimension count. $\square$

When $\beta = 0$ the equation is invariant under the scaling itself, and its waves come in one-parameter families of rates. Order two shows what the orbits mean. There are two of them, and both carry waves. The orbit $\beta = 0$ is the KdV equation, with its family of solitons of all rates. The orbit $\beta \neq 0$ consists of the equations with a nonzero Burgers coefficient. Each of them has one family of fronts, at the rate fixed by that coefficient. The table of Section 9.1 represents this front twice, by the mirror images $a = 1$ and $a = -1$; these are the data (A, λ) and $(\iota A, -\lambda)$ of Proposition 8.1. At $p = 3$, write $P = D^3 + BD^2 + \mu D + a_0$, so that $\beta = (B, \mu)$. The orbits are the curves $B^2 = \sigma \mu$ with the origin removed. For the ten pairs $A = \rho^2 + a\rho + b$ of Table 2, (9.5) gives $B = 4a$ and $\mu = a^2 + 19b$, hence the four classical values

$$\sigma = \frac{16a^2}{a^2 + 19b} \in \left\{0, 16, \frac{144}{47}, \frac{256}{73}\right\}. \quad (8.3)$$

Here the bound $(N_3 + F_3)/2 = 6$ is not attained, because the two pairs with $a = 0$ share $\sigma = 0$, and $(\pm 1, 0)$ and $(\pm i, 0)$ share $\sigma = 16$.

For odd p the classification turns Corollary 8.2 into a statement about all meromorphic travelling waves.

Corollary 8.3 (Generic equations with p odd). *Let $p \geq 5$ be odd and assume rigidity (R_{p-1}) , which holds when $2p - 1$ is prime and for every $p \leq 573$. The coefficient vectors $\beta \in \mathbb{C}^{p-1}$ for which (2.1) has a nonconstant meromorphic travelling wave lie in an algebraic subset of dimension at most two. In particular a generic equation has none. For $p = 3$ this set is the union of the four curves $B^2 = \sigma\mu$ of (8.3).*

Proof. By Theorem 3.1, such a wave is rational, rational in an exponential, or elliptic, with one pole per period. Take $\lambda = 1$ in (2.3), so that $[\rho^j]P = b_j/b_p$ for $1 \leq j \leq p - 1$. The rational-exponential waves give a set of dimension at most one, by Corollary 8.2. For a rational wave (4.1), the argument of Step 3 in the proof of Theorem 6.1 applies verbatim, with the simple-pole term $r_1 z^{-1}$ included: it gives $b = a_0 = 0$, and the Laplace correspondence gives $A \mid H_A$ for a monic A of degree $p - 1$, now without the restriction $A(0) = 0$. By (R_{p-1}) , $A = \rho^{p-1}$, hence $P = D^p$ and $\beta = 0$. For the elliptic waves, the family of all $\mathcal{E}_p(g_2, g_3)$ is finite over the (g_2, g_3) -plane, by Step 4 of the same proof. It has dimension two, so its image under $P \mapsto \beta$ has dimension at most two. For $p = 3$, Section 9.2 gives the elliptic condition $B^2 = 16\mu$, which is one of the four curves; the point $\beta = 0$ lies on all of them. $\square$

The same dimension count applies to the one-pole waves of every order $p \geq 3$ for which (R_{p-1}) holds: a generic equation has no travelling wave that is rational, rational in an exponential, or elliptic with one pole per period. For even p , however, Theorem 3.1 excludes other meromorphic waves only when the equation is purely dispersive.

Other equations with the same profile equation. The reduction of Section 2 uses only the travelling-wave variable. The counts therefore apply to any equation $\mathcal{L}u + \mathcal{M}(u^2) = 0$ with constant-coefficient operators, mixed derivatives and several space variables included, whose plane-wave reduction integrates to (1.1); the coefficients of P then depend on the speed and the direction of the wave. The repository lists such equations by profile operator, with their reductions (directory `tools/wave_dictionary/`).

Ansatz methods. Many exact travelling waves in the literature are produced by substituting a finite expansion: the tanh and sech methods, the exp-function and (G'/G) -expansion methods, the simplest-equation method with a Riccati equation, and the Jacobi and Weierstrass elliptic-function expansions [29]. In their standard form, with one phase $kx - \omega t$ and constant coefficients, the first group produces functions that are rational in one exponential and the second group elliptic functions of one argument; rational functions of z arise as limits. When Theorem 3.1 applies, every such nonconstant meromorphic solution of (2.1) has one pole per period. With respect to its primitive period it is then, up to the symmetries of Proposition 8.1, one of the waves (8.1), one of the elliptic waves of Section 6, or the rational wave. For the remaining operators the same holds for the outputs that have one pole per period. Whenever the corresponding counting hypotheses hold, fixing the exponential rate or the elliptic period lattice bounds the number of normalized outputs by the counts of this paper, independently of the expansion method; changes of rate and background for a fixed evolution equation are governed by Proposition 8.1. For the conformable derivative $T_\alpha f(t) = t^{1-\alpha} f'(t)$, $t > 0$, $0 < \alpha \leq 1$ [25], the change $\tau = t^\alpha/\alpha$ gives $T_\alpha = d/d\tau$, and similarly $X = x^\beta/\beta$ for a spatial conformable derivative; whenever the transformed equation has the profile equation (1.1), its waves in $X - c\tau$ are therefore covered by the same count. This does not apply to the Caputo or Riemann–Liouville derivatives.

9 Examples

The examples have one purpose: to show the counts of Table 1 at work, and to connect them with the classical lists. Orders two and three are small enough for the full family to be treated by hand (Sections 9.1 and 9.2); the second of them also shows what multiplicity means for elliptic pairs. At higher orders we list the pairs of the two pure subfamilies (Sections 9.3 and 9.4). The normalizations are those of Section 2.

9.1 Order two: KdV–Burgers

The integrated travelling-wave equation for

$$u_t + uu_x + \nu u_{xx} + \delta u_{xxx} = 0, \quad \delta \neq 0,$$

has order $p = 2$. Its exact front solutions have been studied in several equivalent forms [22, 29, 47]. The long-wave reductions in Kudryashov et al. [34] also contain this balance of dissipation and dispersion.

Here $A(\rho) = \rho + a$ has a single coefficient, and the matching system is one equation in one unknown. Direct polynomial division gives

$$\text{rem}_A S_A = -\frac{a(a-1)(a+1)}{6}, \quad P_A = \rho^2 + 5a\rho + a^2 - 6a - 1, \quad v_A = 12((a+1)Q - Q^2). \quad (9.1)$$

Thus there are exactly $N_2 = 3$ simple pairs:

a	$P_A(\rho)$	v_A
-1	$(\rho - 2)(\rho - 3)$	$-12Q^2$
0	$\rho^2 - 1$	$12Q(1 - Q) = 3 \operatorname{sech}^2(z/2)$
1	$(\rho - 1)(\rho + 6)$	$12Q(2 - Q)$

(9.2)

The pulse $a = 0$ is the KdV soliton. It is the single pair of the purely dispersive subfamily, $F_2 = 1$. The two fronts belong to mixed equations, and the reflection ι exchanges them: for a given equation with $\nu \neq 0$ they are one and the same front, read in the two directions (Proposition 8.1). At unit rate they are two different pairs, and the count $N_2 = 3$ includes both. In the original variables, (2.3) and (2.4) read

$$u(x, t) = u_* + \delta \kappa^2 v_A(\kappa(x - ct)), \quad \nu = \delta \kappa(5a), \quad u_* - c = \delta \kappa^2(a^2 - 6a - 1).$$

In particular, a nonzero ν fixes the front rate to $\kappa = \nu/(5a\delta)$, $a = \pm 1$, in agreement with the classical parameter restriction.

For a fixed lattice the elliptic candidate is $V = -12X + b$. Its residual is

$$-12a_1Y - 12(a_0 + b)X + 6g_2 + a_0b + \frac{1}{2}b^2.$$

Consequently the complete elliptic system is

$$a_1 = 0, \quad v = -12\wp - a_0, \quad a_0^2 = 12g_2. \quad (9.3)$$

There is no one-pole elliptic member with nonzero Burgers coefficient: both elliptic pairs belong to the KdV equation, in agreement with $M_2 = 2F_2 = 2$. They are the two backgrounds (2.5) of one cnoidal wave. For $g_2 \neq 0$ they are distinct and simple. For $g_2 = 0$, $g_3 \neq 0$, they coincide, and the lattice carries one pair of multiplicity two.

Since $2p - 1 = 3$ is prime, the counts at order two are unconditional. The classification of Theorem 3.1 applies to the KdV subfamily $a_1 = 0$. For the mixed operators, the Fisher example in Section 10 explains why the same conclusion cannot be expected.

9.2 Order three: Kuramoto–Sivashinsky

Let $p = 3$, $s_3 = 60$, and $A(\rho) = \rho^2 + a\rho + b$. The profile and monic operator are

$$\frac{v_A}{60} = 2Q^3 - (a+3)Q^2 + (a+b+1)Q, \quad (9.4)$$

$$P_A(\rho) = \rho^3 + 4a\rho^2 + (a^2 + 19b)\rho - a^3 + 7ab - 5a - 30b. \quad (9.5)$$

The remainder of S_A is $(E_1\rho + E_0)/30$, where

$$E_0 = ab(a^2 - 7b + 5), \quad E_1 = a^4 - 8a^2b + 11b^2 + 10b - 1. \quad (9.6)$$

The system $E_0 = E_1 = 0$ is solved by cases on the factors of E_0 . If $b = 0$, then $a^4 = 1$. If $a = 0$, then $(b+1)(11b-1) = 0$. In the remaining case $a^2 = 7b - 5$, and the second equation becomes $4(b-2)(b-3) = 0$. This exhausts the system and gives Table 2. The Jacobian determinant is nonzero at every point, so all $N_3 = 10$ pairs are simple, as Theorem 5.4 predicts for $2p - 1 = 5$.

Table 2: All ten rational-exponential pairs at order three. The label is the set of roots of A modulo 5 (Theorem 5.4); for the two complex pairs it depends on the choice of i modulo 5. The determinant is taken in the order (E_0, E_1) against (a, b) . The last column describes v_A on the real axis, at unit rate.

a	b	Labels	$\det \partial(E_0, E_1)/\partial(a, b)$	Real profiles
± 1	0	$\{0, 4\}, \{0, 1\}$	-24	Pulses
$\pm i$	0	$\{0, 2\}, \{0, 3\}$	-16	Not real
0	-1	$\{1, 4\}$	144	Front
0	1/11	$\{2, 3\}$	576/121	Front
± 3	2	$\{3, 4\}, \{1, 2\}$	-144	Fronts
± 4	3	$\{2, 4\}, \{1, 3\}$	384	Fronts

The ten labels are the ten two-element subsets of $\{0, 1, 2, 3, 4\}$. The four pulses are the pairs whose label contains 0, and the mirror image $(a, b) \mapsto (-a, b)$ replaces a label S by $-S$, as Proposition 5.6 and Theorem 5.8 assert. There are ten complex pairs, eight with real coefficients. All eight are smooth and bounded on the real axis: two pulses and six fronts. For example,

$$A = (\rho - 1)(\rho - 2), \quad v_A = 120Q^3, \quad P_A = (\rho - 3)(\rho - 4)(\rho - 5).$$

These are the classical solutions in hyperbolic functions [26, 27, 39]. The complete parameter list is tabulated by Kudryashov and Zargaryan [33, Table 1] and Conte and Musette [5, Table 1]. Their tables contain six parameter cases because they identify the two signs of the dispersion coefficient. These six cases are the orbits of the reflection ι , which is $(a, b) \mapsto (-a, b)$ here: four orbits of two pairs, and the two fixed pairs with $a = 0$. Exhaustion of this classical list follows from Eremenko [18].

The two pairs with $a = 0$ are the $F_3 = 2$ pairs of the purely dissipative subfamily, the Kuramoto–Sivashinsky equation without dispersion. Both are fronts, as Theorem 5.8 requires; the pulses $(\pm 1, 0)$ need dispersion. By Corollary 8.2, an equation with $P = D^3 + BD^2 + \mu D + a_0$ has a wave of this kind exactly when $B^2 = \sigma\mu$ for one of the four values (8.3). The pairs $(\pm 1, 0)$ and $(\pm i, 0)$ belong to the same equations, with $\sigma = 16$; their rates differ by a factor i , so for a real equation the second wave is periodic in x , and its real form has poles on the real axis.

Elliptic pairs, and the lattices on which they merge. An elliptic solution must have zero residue at its pole. For $P = D^3 + BD^2 + \mu D + a_0$ the Laurent residue is $60\mu/19 - 15B^2/76$, so $B^2 = 16\mu$. With this restriction the entire elliptic matching scheme is

$$\begin{aligned} P &= D^3 + BD^2 + \frac{B^2}{16}D + a_0, \\ v &= -60\wp' - 15B\wp - \frac{B^3}{64} - a_0, \\ B^4 &= 3072g_2, \quad a_0^2 = \frac{13B^6}{2048} - 2160g_3. \end{aligned} \tag{9.7}$$

Read with the operator fixed, this is the classical solution of Kudryashov [27]: $w = v + a_0$ solves the usual integrated equation $w''' + Bw'' + \mu w' + w^2/2 + K = 0$ with $K = -a_0^2/2$. Necessity of the restriction $B^2 = 16\mu$ is proved by Hone [20], and the classification of all meromorphic KS solutions is completed by Eremenko [18]. For the count we read (9.7) the other way: fix g_2, g_3 and solve for B and a_0 .

For a typical lattice, the equation for B has four solutions, and the equation for a_0 then gives two choices for each of them: the two backgrounds $\pm a_0$ of (2.5). These are the $M_3 = 8$ pairs. When the lattice changes, some of the pairs can meet. The multiplicity of the resulting point records how many nearby pairs have come together, so the count with multiplicity can remain eight even when fewer than eight distinct pairs remain.

Pairs meet exactly where the Jacobian of the last two equations of (9.7) vanishes, and that Jacobian is a nonzero constant times $B^3 a_0$. Let

$$j(\Lambda) = \frac{1728g_2^3}{g_2^3 - 27g_3^2}.$$

If $B = 0$, then $g_2 = 0$, hence $j = 0$. If $a_0 = 0$ and the lattice is nonsingular, elimination gives

$$675g_3^2 = 169g_2^3, \quad j = -300.$$

These conditions are sufficient as well. Thus:

$j(\Lambda)$	distinct pairs	total with multiplicity
$j \notin \{0, -300\}$	8	8×1
-300	6	$2 \times 2 + 4 \times 1$
0	2	2×4 .

(9.8)

The two exceptional cases show the two ways in which pairs can meet.

At $j = -300$, the four values of B remain distinct. For two of them, the two choices of a_0 meet at zero: the two backgrounds of one wave coincide. Each produces one pair of multiplicity two, while the other four pairs remain distinct. There are six distinct pairs, with total multiplicity $2 + 2 + 1 + 1 + 1 + 1 = 8$. For instance, $g_2 = 1/12$, $g_3 = 13/1080$ gives the double points $(B, a_0) = (\pm 4, 0)$.

At $j = 0$, all four values of B meet at $B = 0$. The two roots of $a_0^2 = -2160g_3$ remain distinct, since $g_3 \neq 0$, so there are two pairs, each of multiplicity four. Both have the operator $D^3 + a_0$, which is purely dissipative, and the total is again $4 + 4 = 8$.

The count stays eight for an exact reason. The coefficients eliminated on the way to (9.7) are determined linearly, with nonzero rational pivots, so its last two equations carry the full matching system, multiplicities included. The first has four roots B and, for each of them, the second has two roots a_0 , counted with multiplicity. This gives eight solutions on every lattice. This is the form in which Theorem 6.1 holds on an arbitrary lattice, and (9.8) is the exceptional set of that theorem at order three.

9.3 Purely dispersive equations: Kawahara and seventh-order KdV

Order four. For the Kawahara equation [14, 23], the integrated, shifted and monic travelling-wave operator is

$$P = D^4 + bD^2 + a_0. \quad (9.9)$$

The classification for this equation, using its first integral, is already established by Demina and Kudryashov [14]; see also Ye et al. [48]. Their first integral is the order-four instance of (3.7).

By Theorem 5.8, the pairs with an even operator are those with $\iota A = A$, that is, $A = \rho^3 + c\rho$. All of them are pulses: the profile is even about its pole, $v(Q) = v(1 - Q)$. The quotient and remainder are

$$\begin{aligned} P_A &= \rho^4 + 13c\rho^2 + \frac{31c^2 - 70c + 7}{3}, \\ v_A &= 280(1 + c)Q(1 - Q) - 1680Q^2(1 - Q)^2, \\ \text{rem}_A S_A &= -\frac{\rho}{420}(c + 1)(31c^2 - 31c + 10). \end{aligned} \quad (9.10)$$

The three distinct roots are

$$c = -1, \quad c = \frac{1}{2} \pm \frac{3i}{2\sqrt{31}}.$$

They give $F_4 = \binom{3}{1} = 3$ simple pairs. These are the three simply periodic parameter choices of Demina and Kudryashov [14, Section 3], in the present normalization: the parameter in their Eq. (31) equals $1/c$ here. At $c = -1$,

$$v = -105 \operatorname{sech}^4(z/2), \quad P = (D^2 - 4)(D^2 - 9). \quad (9.11)$$

This is the familiar solitary wave of the Kawahara equation; the other two pairs are complex at unit real rate. The unrestricted order-four count is $N_4 = 35$. Of these pairs, 21 have real coefficients, and all 21 are rational: five pulses and sixteen fronts, which are eleven wave shapes up to reflection. They are the one-pole waves of the mixed fifth-order equations, such as the Benney–Lin equation, and the repository dictionary lists them.

All elliptic members of (9.9) have the form

$$v = -1680\wp^2 - \frac{280b}{13}\wp + 168g_2 + \frac{31b^2}{507} - a_0. \quad (9.12)$$

Substitution gives exactly two conditions:

$$F(b; g_2, g_3) := 31b^3 - 42588bg_2 - 4745520g_3 = 0, \quad (9.13)$$

$$a_0^2 = 23184g_2^2 + \frac{7980}{169}b^2g_2 + \frac{101520}{13}bg_3. \quad (9.14)$$

These are the Weierstrass profiles and parameter relations obtained from Laurent series by Demina and Kudryashov [14, Section 3], expressed in the normalization (1.1). In their notation, $b = -\beta$, $a_0^2 = C_0^2 + 12C_1$; the profile change is $v = -6(w - w_*)$, where w_* is a constant equilibrium and $a_0 = C_0 - 6w_*$. The first equation is cubic in b , and the second is monic quadratic in a_0 , so there are $3 \times 2 = 6$ solutions, counted with multiplicity, on every fixed lattice: three values of b , each with its two backgrounds. This is the number $2F_4$ of Corollary 6.2. At $(g_2, g_3) = (1, 0)$, the cubic has three distinct roots and the right side of (9.14) is nonzero at all of them, so the six pairs are simple on a nonempty open set of lattices. The unrestricted elliptic count at this order is $M_4 = 30$; elliptic formulas with nonzero odd derivative coefficients in the full fifth-order evolution equation are given by Kudryashov and Sinelshchikov [37].

Order six. The seventh-order KdV equation has the operator $P = D^6 + a_4 D^4 + a_2 D^2 + a_0$; its meromorphic travelling waves are studied by Dang [9]. Here $A = \rho^5 + c_1 \rho^3 + c_2 \rho$ and $s_6 = 5544$. The remainder of S_A has two coefficients, at ρ^3 and at ρ , which are polynomials in c_1, c_2 of weights eight and ten; the ancillary script prints them. Eliminating c_2 , which is determined by c_1 at every solution, gives

$$(c_1 + 5)(c_1 + 10)(49c_1^2 - 5c_1 + 10)g_6(c_1) = 0, \quad (9.15)$$

where g_6 is an irreducible sextic without real roots, also printed by the script. The ten roots are distinct, in agreement with $F_6 = \binom{5}{2} = 10$. Two pairs are real:

A	P_A	v_A
$\rho(\rho^2 - 1)(\rho^2 - 4)$	$(\rho^2 - 9)(\rho^2 - 16)(\rho^2 - 25)$	$10395 \operatorname{sech}^6(z/2)$
$\rho(\rho^2 - 1)(\rho^2 - 9)$	$(\rho^2 - 4)(\rho^2 - 25)(\rho^2 - 71)$	$10395 \operatorname{sech}^4(z/2)(1 + \operatorname{sech}^2(z/2))$.

(9.16)

All ten pairs are pulses. An exact computation on a sample lattice gives $2F_6 = 20$ for the elliptic pairs of this family.

9.4 Purely dissipative equations of order five

At $p = 5$ the purely dissipative operators are $P = D^5 + a_3 D^3 + a_1 D + a_0$, and $A = \rho^4 + c_1 \rho^2 + c_2$. All 126 pairs at $p = 5$ are simple (Section 7), so there are exactly $F_5 = 6$ distinct pairs here. They are

$$A = (\rho^2 - 1)(\rho^2 - 4), \quad A = (\rho^2 - 1)(\rho^2 - 9), \quad A = (\rho^2 - 1)\left(\rho^2 + \frac{1}{5}\right), \quad (9.17)$$

and the three pairs with

$$1105c_1^3 - 2969c_1^2 + 5570c_1 - 4000 = 0, \quad c_2 = \frac{5c_1^3 + 21c_1 - 10}{79c_1 + 210}.$$

The cubic has one real root. All six pairs are fronts, as Theorem 5.8 requires, and the four with real coefficients are, after the scales are restored, the solutions (1.4) of Kudryashov and Migita [32]. For instance, $A = (\rho^2 - 1)(\rho^2 - 4)$ gives

$$v_A = 5040 Q^3(10 - 15Q + 6Q^2), \quad P_A = \rho^5 - 55\rho^3 + 1254\rho - 2520,$$

a front from 0 to $5040 = -2P_A(0)$.

10 Open questions and the boundary of the classification

Two questions remain open in the enumeration, and one limitation of the classification deserves a precise statement.

Rigidity. The first question is the continuous-convolution rigidity assertion

$$0 \neq A \in \mathbb{C}[\rho], \quad A \mid \int_0^\rho A(t)A(\rho - t) dt \implies A = c\rho^{\deg A}.$$

Through the polynomial Laplace correspondence used in the proof of Theorem 6.1, it says that, after fixing the pole at zero, any rational one-pole solution of (1.1) vanishing at infinity would have to be $c_* z^{-p}$, with operator $P = D^p$. Algebraic and ℓ -adic arguments, together with exact computations, prove it in the range $\deg A \leq 572$, corresponding to $2 \leq p \leq 573$, and in infinite arithmetic families of higher degrees. Rigidity in arbitrary degree remains a conjecture. By Proposition 7.1, it would establish both enumeration formulas, as counts with multiplicity, at every order.

Simplicity. The second question is whether every point of $\mathcal{X}_p$, the rational-exponential matching system, is simple. The elliptic simplicity assertion is generic in the lattice and is derived from this one in Theorem 6.1. At a solution $S_A = AC_A$, the tangent map of the system is

$$J_A(h) = \text{rem}_A(2(A \star h) - C_A h), \quad \deg h < d. \quad (10.1)$$

The question is whether this map always has zero kernel. The prime-index argument proves this after reduction modulo ℓ , where the map becomes multiplication by a polynomial coprime to $\bar{A}$; a uniform replacement is missing at composite indices. One might expect simplicity to follow from the uniqueness of the Laurent expansion in Theorem 3.1. It does not: Laurent uniqueness concerns a fixed operator, whereas (10.1) allows the operator to vary with the profile.

Conjecture 10.1. Continuous-convolution rigidity holds in every degree, and the matching Jacobian (10.1) is invertible at every point of $\mathcal{X}_p$. Consequently there are exactly $\binom{2p-1}{p-1}$ rational-exponential pairs, all distinct, for every $p \geq 2$.

The conjecture would also imply M_p simple elliptic pairs on a generic fixed lattice at every order, F_p distinct pairs in the two pure subfamilies, and that the rational-exponential waves of every purely dissipative equation are fronts (Theorem 5.8). Simplicity on every lattice is false already at order two; (9.8) describes the full exceptional set at order three.

Mixed equations with p even. The hypothesis of Theorem 3.1 is a separate issue. For even p it requires a purely dispersive equation, and it cannot simply be dropped. The known example comes from the Fisher equation. Its travelling waves of nonzero speed satisfy (1.1) with $p = 2$ and a mixed operator, which is also the profile equation of KdV–Burgers. At the special speed of Ablowitz and Zeppetella [1] it has a one-parameter family of solutions, given in Kudryashov [30], Conte et al. [6, Lemma 4.4] and McCue et al. [42]; see also Kudryashov [28]. In our normalization it is

$$v(z) = -12\lambda^2 e^{2\lambda z} \wp(e^{\lambda z} - 1; 0, g_3), \quad P(D) = D^2 - 5\lambda D + 6\lambda^2, \quad \lambda \neq 0.$$

Substitution follows from $(D^2 - 5\lambda D + 6\lambda^2)[t^2 U(t)] = \lambda^2 t^4 U''(t)$, $t = e^{\lambda z}$. For $g_3 = 0$ and $\lambda = 1$ this is the front in the first row of (9.2). For generic $g_3 \neq 0$ it is globally meromorphic and lies outside $\mathcal{W}$. Further examples of meromorphic solutions outside $\mathcal{W}$ are discussed by Ng and Wu [43, Remark 1.11]. Thus a classification for mixed operators with p even must allow a larger class of functions than $\mathcal{W}$.

For mixed operators of even degree $p \geq 4$ we do not know whether meromorphic solutions outside $\mathcal{W}$ occur. Methods for constructing elliptic and rational-exponential solutions without the finiteness property are developed by Demina [13].

A Exact verification and ancillary material

This appendix describes the computer checks that accompany the paper, and the companion note.

The companion note `anc/general_orders.pdf` contains what is known at the orders for which $2p - 1$ is not prime: the weighted Bézout count behind Proposition 7.1, the general-order versions of the results of Sections 5.3, 5.4 and 6, and the proof of rigidity when $2p - 1$ is a prime power.

The ancillary material contains five kinds of checks.

1. `verify_manuscript.py` reconstructs the discrete and continuous convolutions, the low-order Laurent coefficients, and the Weierstrass differential algebra. It also checks the leading-term comparison used in Proposition 5.2. It checks the displayed order-two, order-three and order-four formulas by exact substitution, including the correspondence with the KS tables of Conte and Musette [5], Kudryashov and Zargaryan [33], the KS collision multiplicities, the Kawahara coefficient restriction and its literature normalization, the nodal correspondence, and the Fisher identity. These finite substitutions check the displayed formulas; the uniform proofs are the arguments in the text.
2. `p5_count.sing` is a self-contained rational SINGULAR calculation for the full order-five matching ideal. Its quotient dimension is 126. Adjoining the matching Jacobian determinant gives the unit ideal, so all 126 points are simple. The input is reconstructed from the original polynomial sum by the Python checker; the saved transcript states the two exact outputs.
3. At $p = 8$ the same two outputs, 6435 and the unit ideal, are obtained over the finite field $\mathbb{F}_{32003}$. Every point modulo 32003 is therefore simple and lifts uniquely by Hensel's lemma, as in Step 4 of the proof of Theorem 5.4, and the lifts are all the points because the count with multiplicity is 6435, by Proposition 7.1(ii). At $p = 11$ each of the 352716 points is certified over the 19-adic numbers, by Hensel's lemma at the simple points of the reduction modulo 19 and by exact Newton-polygon and Newton-Kantorovich certificates at the singular ones; the archived certificate, with its description and the record of its validation, is in the repository, directory `certificates/`.
4. `rigidity/` supplies the proof and exact certificate for rigidity (R_d) , $d \leq 72$. For $2d + 1 = \ell^e + g$, $g \in \{0, 2, 4, 6\}$, the proof reduces the normalized continuous convolution to a residual system of degree at most g . The certificate retains every residual degree, with integer ideal-membership identities and certified prime factors. The replay reconstructs the inputs, multiplies out every identity, and checks all 72 arithmetic choices. It does not infer characteristic-zero nonexistence from a list of residue-field points.
5. `symmetry_checks/` contains the exact checks for the two pure subfamilies and for single equations: the identities of Proposition 5.7 with symbolic coefficients for $d \leq 5$; the labels of Table 2 and of the 21 real pairs at $p = 4$, with the statements of Proposition 5.6 and Theorem 5.8(ii) about them; the lengths F_p for $p \leq 8$, and finite-field certificates, of the same kind as at $p = 8$, that the F_p pairs are distinct for every $p \leq 14$; the lengths $2F_p$ of Corollary 6.2 for $p \leq 6$ on a sample lattice; the six pairs (9.17) and the ten pairs (9.15), with the sextic factor in full; and the scaling (8.1) by substitution in the evolution equation.

A Lean 4 formalization accompanies the paper at <https://github.com/migita/binomial-count-meromorphic-waves>, directory `lean/`, with Lean and Mathlib pinned to version 4.33.1. It proves (R_d) for $2d + 1$ prime (`rigidity_prime_index`) and the count and simplicity statements of Theorem 5.4, both for the matching system (5.6) (`primeIndexCountAndSimplicity`) and for the normalized pairs (P, v) as solutions of (1.1) in the field of rational functions of e^z (`primeIndexRationalExponentialCountAndSimplicity`), together with the correspondence between the two formulations for every $p \geq 2$. The discrete convolution is defined there by the Bernoulli-polynomial power sums, and the remainder equations and their Jacobian are those of Section 5. The proof route is outlined in Remark 5.5 and documented in the repository. The files `EnumerationCheck.lean` and `Check.lean` verify that the proofs have

exactly the stated propositions as their types and that the theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; the project contains no `sorry`, added axiom or `native_decide`.

The scripts use exact rational or finite-field arithmetic. Instructions, dependency versions and checksums are included with the source package. The prime-index theorem does not depend on the computer calculations.

Tool and computational resource disclosure

This paper was written with substantial use of generative AI, which is disclosed here in the form recommended by the Leiden Declaration on Artificial Intelligence and Mathematics¹ and required by arXiv. No automated system is an author; credit and responsibility for the results, and for the attribution of prior work, rest with the author.

- **gpt-6-astra** (OpenAI), August–September 2026. The main mathematical ideas of the paper, the proofs of Theorems 3.1, 5.4 and 6.1 and of the weighted count in the companion note, the rigidity condition (R_d) and its certificate, the exact verification scripts, the SINGULAR input, the first complete draft of the text, and the Lean 4 formalization of Remark 5.5 were produced by this model in dialogue with the author.
- OpenAI Codex v0.153.4, September 2026. Independent mathematical audit of the final revision, integration of the edits, and rebuilding and validation of the source package.
- Claude Code (Anthropic, Claude Fable 5.1), September 2026. The literature search behind the attributions in Section 1, the readability revision of all sections, and the leading-coefficient argument of Proposition 5.2, which Codex then checked independently. In the final revision it proposed and proved the results on the two pure subfamilies and on single equations (Corollary 3.2, Section 5.4, Corollary 6.2 and Section 8), with their exact checks, and reorganized the manuscript around them. In the last revision it made the prime-index theorem the spine of the paper: it wrote the proof of Theorem 5.4 in the form used by the Lean formalization, derived Proposition 5.6 and Theorem 5.8 from the labels, replaced the Bézout count in the proof of Theorem 6.1 by a flat-family argument, and moved the general-order material to the companion note; Codex reviewed these arguments. It also audited the Lean formalization independently: the formal statements against the manuscript, a rebuild from source, and the axiom audit.

References

- [1] Mark J. Ablowitz and Anthony Zeppetella. Explicit solutions of Fisher’s equation for a special wave speed. *Bulletin of Mathematical Biology*, 41(6):835–840, 1979. doi:[10.1007/BF02462380](https://doi.org/10.1007/BF02462380).
- [2] D. J. Benney. Long waves on liquid films. *Journal of Mathematics and Physics*, 45:150–155, 1966. doi:[10.1002/sapm1966451150](https://doi.org/10.1002/sapm1966451150).
- [3] Keith Conrad. A multivariable Hensel’s lemma. University of Connecticut, expository notes, n.d. URL <https://kconrad.math.uconn.edu/blurbs/gradnumthy/multivarhensel.pdf>. Theorem 3.3.

¹<https://leidendeclaration.ai/>, 2 June 2026.

- [4] Robert Conte and Micheline Musette. The Painlevé methods. In A. Kundu, editor, *Classical and Quantum Nonlinear Integrable Systems: Theory and Application*, pages 39–63. IOP Publishing, Bristol, 2003. doi:[10.1201/9781420034615.ch2](https://arxiv.org/abs/nlin/0211048). URL <https://arxiv.org/abs/nlin/0211048>.
- [5] Robert Conte and Micheline Musette. Elliptic general analytic solutions. *Studies in Applied Mathematics*, 123(1):63–81, 2009. doi:[10.1111/j.1467-9590.2009.00447.x](https://arxiv.org/abs/1467-9590.2009.00447.x).
- [6] Robert Conte, Tuen-Wai Ng, and Cheng-Fa Wu. Hayman’s classical conjecture on some nonlinear second-order algebraic ODEs. *Complex Variables and Elliptic Equations*, 60(11):1539–1552, 2015. doi:[10.1080/17476933.2015.1033414](https://arxiv.org/abs/1503.07074). URL <https://arxiv.org/abs/1503.07074>.
- [7] Robert Conte, Tuen Wai Ng, and Chengfa Wu. Singularity methods for meromorphic solutions of differential equations. In N. Euler, editor, *Nonlinear Systems and Their Remarkable Mathematical Structures*, volume 1, pages 159–186. CRC Press, Boca Raton, 2018. URL <https://arxiv.org/abs/1806.03177>.
- [8] Robert Conte, Micheline Musette, Tuen Wai Ng, and Chengfa Wu. Methods for exact solutions of nonlinear ordinary differential equations. *Advances in Mathematics (China)*, 54(2):379–389, 2025. doi:[10.11845/sxjz.2024010a](https://arxiv.org/abs/2503.00834). URL <https://arxiv.org/abs/2503.00834>.
- [9] Guoqiang Dang. Meromorphic solutions of the seventh-order KdV equation by using an extended complex method and Painlevé analysis. *ScienceAsia*, 49(1):108–115, 2023. doi:[10.2306/scienceasia1513-1874.2022.133](https://arxiv.org/abs/2306.1513-1874.2022.133).
- [10] Guoqiang Dang. Elliptic and multiple-valued solutions of some higher order ordinary differential equations. *Electronic Research Archive*, 31(10):5946–5958, 2023. doi:[10.3934/era.2023302](https://arxiv.org/abs/2303.3934-era.2023302).
- [11] M. V. Demina and N. A. Kudryashov. Doubly periodic meromorphic solutions of autonomous nonlinear differential equations. *Automatic Control and Computer Sciences*, 48(7):633–641, 2014. doi:[10.3103/S0146411614070207](https://arxiv.org/abs/10.3103/S0146411614070207). Russian original: Modeling and Analysis of Information Systems 21(5) (2014), 49–60.
- [12] Maria V. Demina. Classification of meromorphic integrals for autonomous nonlinear ordinary differential equations with two dominant monomials. *Journal of Mathematical Analysis and Applications*, 479(2):1851–1862, 2019. doi:[10.1016/j.jmaa.2019.07.026](https://arxiv.org/abs/10.1016/j.jmaa.2019.07.026).
- [13] Maria V. Demina. Meromorphic solutions of autonomous ordinary differential equations without the finiteness property. *Journal of Mathematical Analysis and Applications*, 516(2):126516, 2022. doi:[10.1016/j.jmaa.2022.126516](https://arxiv.org/abs/10.1016/j.jmaa.2022.126516).
- [14] Maria V. Demina and Nikolay A. Kudryashov. From Laurent series to exact meromorphic solutions: The Kawahara equation. *Physics Letters A*, 374(39):4023–4029, 2010. doi:[10.1016/j.physleta.2010.08.013](https://arxiv.org/abs/10.1016/j.physleta.2010.08.013).
- [15] Maria V. Demina and Nikolay A. Kudryashov. Explicit expressions for meromorphic solutions of autonomous nonlinear ordinary differential equations. *Communications in Nonlinear Science and Numerical Simulation*, 16(3):1127–1134, 2011. doi:[10.1016/j.cnsns.2010.06.035](https://arxiv.org/abs/10.1016/j.cnsns.2010.06.035).
- [16] Maria V. Demina and Nikolay A. Kudryashov. On elliptic solutions of nonlinear ordinary differential equations. *Applied Mathematics and Computation*, 217(23):9849–9853, 2011. doi:[10.1016/j.amc.2011.04.043](https://arxiv.org/abs/10.1016/j.amc.2011.04.043).
- [17] A. E. Eremenko. Meromorphic solutions of algebraic differential equations. *Russian Mathematical Surveys*, 37(4):61–95, 1982. doi:[10.1070/RM1982v037n04ABEH003967](https://arxiv.org/abs/10.1070/RM1982v037n04ABEH003967). Erratum: Russian Mathematical Surveys 38(6) (1983).
- [18] Alexandre Eremenko. Meromorphic traveling wave solutions of the Kuramoto–Sivashinsky equation. *Journal of Mathematical Physics, Analysis, Geometry*, 2(3):278–286, 2006. URL <https://arxiv.org/abs/nlin/0504053>.
- [19] Alexandre E. Eremenko, Liangwen Liao, and Tuen Wai Ng. Meromorphic solutions of higher order Briot–Bouquet differential equations. *Mathematical Proceedings of the Cambridge Philosophical Society*, 146(1):197–206, 2009. doi:[10.1017/S030500410800176X](https://arxiv.org/abs/10.1017/S030500410800176X). URL <https://arxiv.org/abs/0707.2898>.

- [20] A. N. W. Hone. Non-existence of elliptic travelling wave solutions of the complex Ginzburg–Landau equation. *Physica D: Nonlinear Phenomena*, 205(1–4):292–306, 2005. doi:[10.1016/j.physd.2004.10.011](https://doi.org/10.1016/j.physd.2004.10.011).
- [21] Zifeng Huang, Liming Zhang, Qiuhui Chen, and Wenjun Yuan. The representation of meromorphic solutions for a class of odd order algebraic differential equations and its applications. *Mathematical Methods in the Applied Sciences*, 37(10):1553–1560, 2014. doi:[10.1002/mma.3053](https://doi.org/10.1002/mma.3053).
- [22] Alan Jeffrey and Siqing Xu. Exact solutions to the Korteweg–de Vries–Burgers equation. *Wave Motion*, 11(6):559–564, 1989. doi:[10.1016/0165-2125\(89\)90026-7](https://doi.org/10.1016/0165-2125(89)90026-7).
- [23] Takuji Kawahara. Oscillatory solitary waves in dispersive media. *Journal of the Physical Society of Japan*, 33(1):260–264, 1972. doi:[10.1143/JPSJ.33.260](https://doi.org/10.1143/JPSJ.33.260).
- [24] Takuji Kawahara and Sadayoshi Toh. Pulse interactions in an unstable dissipative-dispersive nonlinear system. *Physics of Fluids*, 31:2103–2111, 1988. doi:[10.1063/1.866610](https://doi.org/10.1063/1.866610).
- [25] R. Khalil, M. Al Horani, A. Yousef, and M. Sababheh. A new definition of fractional derivative. *Journal of Computational and Applied Mathematics*, 264:65–70, 2014. doi:[10.1016/j.cam.2014.01.002](https://doi.org/10.1016/j.cam.2014.01.002).
- [26] N. A. Kudryashov. Exact soliton solutions of the generalized evolution equation of wave dynamics. *Journal of Applied Mathematics and Mechanics*, 52(3):361–365, 1988. doi:[10.1016/0021-8928\(88\)90090-1](https://doi.org/10.1016/0021-8928(88)90090-1).
- [27] N. A. Kudryashov. Exact solutions of the generalized Kuramoto–Sivashinsky equation. *Physics Letters A*, 147(5–6):287–291, 1990. doi:[10.1016/0375-9601\(90\)90449-X](https://doi.org/10.1016/0375-9601(90)90449-X).
- [28] N. A. Kudryashov. Exact solitary waves of the Fisher equation. *Physics Letters A*, 342(1–2):99–106, 2005. doi:[10.1016/j.physleta.2005.05.025](https://doi.org/10.1016/j.physleta.2005.05.025).
- [29] N. A. Kudryashov. Seven common errors in finding exact solutions of nonlinear differential equations. *Communications in Nonlinear Science and Numerical Simulation*, 14(9–10):3507–3529, 2009. doi:[10.1016/j.cnsns.2009.01.023](https://doi.org/10.1016/j.cnsns.2009.01.023).
- [30] N. A. Kudryashov. Meromorphic solutions of nonlinear ordinary differential equations. *Communications in Nonlinear Science and Numerical Simulation*, 15(10):2778–2790, 2010. doi:[10.1016/j.cnsns.2009.11.013](https://doi.org/10.1016/j.cnsns.2009.11.013). URL <https://arxiv.org/abs/1201.0126>.
- [31] N. A. Kudryashov. Method of the logistic function for finding analytical solutions of nonlinear differential equations. *Modeling and Analysis of Information Systems*, 22(1):23–37, 2015. doi:[10.18255/1818-1015-2015-1-23-37](https://doi.org/10.18255/1818-1015-2015-1-23-37). URL <https://www.mathnet.ru/eng/mais426>.
- [32] N. A. Kudryashov and A. V. Migita. Periodic structures developing with account for dispersion in a turbulence model. *Fluid Dynamics*, 42(3):463–471, 2007. doi:[10.1134/S0015462807030143](https://doi.org/10.1134/S0015462807030143).
- [33] N. A. Kudryashov and E. D. Zargaryan. Solitary waves in active-dissipative dispersive media. *Journal of Physics A: Mathematical and General*, 29(24):8067–8077, 1996. doi:[10.1088/0305-4470/29/24/029](https://doi.org/10.1088/0305-4470/29/24/029).
- [34] N. A. Kudryashov, D. I. Sinelshchikov, and I. L. Chernyavsky. Nonlinear evolution equations for description of perturbations in a viscoelastic tube. *Russian Journal of Nonlinear Dynamics*, 4(1): 69–86, 2008. doi:[10.20537/nd0801004](https://doi.org/10.20537/nd0801004). In Russian.
- [35] Nikolay A. Kudryashov. One method for finding exact solutions of nonlinear differential equations. *Communications in Nonlinear Science and Numerical Simulation*, 17(6):2248–2253, 2012. doi:[10.1016/j.cnsns.2011.10.016](https://doi.org/10.1016/j.cnsns.2011.10.016).
- [36] Nikolay A. Kudryashov. Polynomials in logistic function and solitary waves of nonlinear differential equations. *Applied Mathematics and Computation*, 219(17):9245–9253, 2013. doi:[10.1016/j.amc.2013.03.032](https://doi.org/10.1016/j.amc.2013.03.032).
- [37] Nikolay A. Kudryashov and Dmitry I. Sinelshchikov. Elliptic solutions for a family of fifth order nonlinear evolution equations. *Applied Mathematics and Computation*, 218(12):6991–6997, 2012. doi:[10.1016/j.amc.2011.12.058](https://doi.org/10.1016/j.amc.2011.12.058).

- [38] Nikolay A. Kudryashov, Dmitry I. Sinelshchikov, and Maria V. Demina. Exact solutions of the generalized Bretherton equation. *Physics Letters A*, 375(7):1074–1079, 2011. doi:[10.1016/j.physleta.2011.01.010](https://doi.org/10.1016/j.physleta.2011.01.010).
- [39] Y. Kuramoto and T. Tsuzuki. Persistent propagation of concentration waves in dissipative media far from thermal equilibrium. *Progress of Theoretical Physics*, 55(2):356–369, 1976. doi:[10.1143/PTP.55.356](https://doi.org/10.1143/PTP.55.356).
- [40] S. P. Lin. Finite amplitude side-band stability of a viscous film. *Journal of Fluid Mechanics*, 63: 417–429, 1974. doi:[10.1017/S0022112074001704](https://doi.org/10.1017/S0022112074001704).
- [41] H. Matsumura. *Commutative Ring Theory*, volume 8 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1986. doi:[10.1017/CBO9781139171762](https://doi.org/10.1017/CBO9781139171762).
- [42] Scott W. McCue, Maud El-Hachem, and Matthew J. Simpson. Exact sharp-fronted travelling wave solutions of the Fisher–KPP equation. *Applied Mathematics Letters*, 114:106918, 2021. doi:[10.1016/j.aml.2020.106918](https://doi.org/10.1016/j.aml.2020.106918).
- [43] Tuen-Wai Ng and Cheng-Fa Wu. Nonlinear Loewy factorizable algebraic ODEs and Hayman’s conjecture. *Israel Journal of Mathematics*, 229(1):1–38, 2019. doi:[10.1007/s11856-018-1791-0](https://doi.org/10.1007/s11856-018-1791-0). URL <https://arxiv.org/abs/1710.08593>.
- [44] V. N. Nikolaevskii. Dynamics of viscoelastic media with internal oscillators. In *Recent Advances in Engineering Science*, volume 39 of *Lecture Notes in Engineering*, pages 210–221. Springer, Berlin, 1989. doi:[10.1007/978-3-642-83695-4_21](https://doi.org/10.1007/978-3-642-83695-4_21).
- [45] NIST. Digital library of mathematical functions, n.d. URL <https://dlmf.nist.gov/>. Chapters 22 and 23.
- [46] Eman Simbawa, Paul C. Matthews, and Stephen M. Cox. Nikolaevskiy equation with dispersion. *Physical Review E*, 81(3):036220, 2010. doi:[10.1103/PhysRevE.81.036220](https://doi.org/10.1103/PhysRevE.81.036220).
- [47] M. Vlieg-Hulstman and W. D. Halford. The Korteweg–de Vries–Burgers’ equation: a reconstruction of exact solutions. *Wave Motion*, 14(3):267–271, 1991. doi:[10.1016/0165-2125\(91\)90041-L](https://doi.org/10.1016/0165-2125(91)90041-L).
- [48] Feng Ye, Jian Tian, Xiaoting Zhang, Chunling Jiang, Tong Ouyang, and Yongyi Gu. All traveling wave exact solutions of the Kawahara equation using the complex method. *Axioms*, 11(7):330, 2022. doi:[10.3390/axioms11070330](https://doi.org/10.3390/axioms11070330).
- [49] Wenjun Yuan, Yezhou Li, and Jianming Qi. All meromorphic solutions of some algebraic differential equations and their applications. *Advances in Difference Equations*, 2014:105, 2014. doi:[10.1186/1687-1847-2014-105](https://doi.org/10.1186/1687-1847-2014-105).
- [50] Wenjun Yuan, Yonghong Wu, Qiuhui Chen, and Yong Huang. All meromorphic solutions for two forms of odd order algebraic differential equations and its applications. *Applied Mathematics and Computation*, 240:240–251, 2014. doi:[10.1016/j.amc.2014.04.099](https://doi.org/10.1016/j.amc.2014.04.099).